The Physics in Economics

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*Dedicated to Dr. Yetter for the **energy** brought into every class and sparking my interest in Economics for the very first time in my life.*

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1 Introduction

When I was watching the lectures on Market Equilibrium, I noticed a certain term being used: "Market Forces". That stuck with me for a while. As an Engineer and a Mathematician, it sparked the freshman curiosity in me, and I began to wonder if a market could be treated as a dynamical system. It made a lot of sense to say that it was true: two opposing wants of the consumer and the supplier negotiating to come to a stable price resembles two opposing effects driving a system toward equilibrium.

The purpose of this paper is not to claim that a market is literally a mechanical body. Rather, the goal is to show that under idealized assumptions, the behaviour of a market can be studied using the same mathematical structure as a Physical System. In particular, the focus is on the behaviour of a market under *Ceteris Paribus*, where all external factors are held constant and Demand and Supply are treated as functions of Price/Quantity alone. If one additionally assumes perfect information, instantaneous adjustment, and absence of transaction or information delays, then there are no explicit mechanisms that would naturally produce lagged overshooting or oscillatory behaviour.

This makes the idealized market resemble a First-Order relaxation system: a system whose state moves directly and monotonically toward equilibrium. A canonical Physical example of such a system is Free Fall with Linear Drag. In Free Fall with Linear Drag, the body does not oscillate around terminal velocity. Instead, it approaches terminal velocity smoothly because the net acceleration decreases as velocity increases. Similarly, in an idealized market, the net imbalance between Demand and Supply decreases as the market moves closer to equilibrium.

Therefore, Free Fall is used in this paper not merely because the equations look similar, but because the qualitative dynamics are similar: both systems involve a net imbalance, a stable equilibrium, no oscillation, and convergence toward that equilibrium through a First-Order process.

2 The Connection

The connection between Physics and Economics is undeniable, especially considering the fact that the Market Price can actually be mathematically modeled as a function of the number of Goods produced. Upon pondering for a long time and pacing countless steps in my room, I saw a strong connection, rooted in intuition and mathematical reasoning, between the Economic Market and the well known Newton's Second Law, which states that the net force is the vector/algebraic sum of all forces acting on a body.

In the case of Free Fall with Linear Drag, a constant driving force is opposed by a linear resistive force. The result is a First-Order Relaxation system that converges toward terminal velocity. This convergence is not oscillatory because the system does not contain a second-order restoring mechanism or a delayed feedback term. The net force simply decreases until it vanishes at terminal velocity.

A Market under *Ceteris Paribus* can be viewed similarly. When Demand exceeds Supply, there is upward market pressure. When Supply exceeds Demand, there is downward market pressure. When Demand equals Supply, the net market pressure vanishes and equilibrium is reached. If traders are assumed to be perfectly informed, if adjustment is assumed to be instantaneous, and if information or transaction delays are ignored, then the model has no reason to include oscillatory or lagged behaviour. The market can therefore be treated as a First-Order system moving toward equilibrium.

This is the central reason for using Free Fall as the Physical analogue. The analogy is not

based only on the visual resemblance between two mathematical expressions. It is based on the fact that both the idealized market and Free Fall with Linear Drag exhibit the same kind of relaxation behaviour. In the case of *Ceteris Paribus* where Supply and Demand Curves are linear, the Market can be specifically associated with the Physical System of an Object in Free Fall influenced by a Drag Force.

3 Mathematical Resemblance

The mathematical similarity between Market and Free Fall Dynamics comes from their shared First-Order relaxation structure.

Market: In a Market with linear Supply and Demand curves, only two functions are present — a Supply function and a Demand function. The equilibrium is simply the intersection point between the two functions. Mathematically:

$$\left.\begin{aligned} P_{supply} &= a + bQ \\ P_{demand} &= c - dQ \end{aligned}\right\} a, b, c, d \in \mathbb{R}^+$$

To find intersection:

$$\begin{aligned} P_{supply} &= P_{demand} \\ \Rightarrow a + bQ &= c - dQ \\ \Rightarrow 0 &= c - dQ - a - bQ \\ \Rightarrow 0 &= c - a - (d + b)Q \end{aligned} \tag{1}$$

The RHS of equation 1 is $c - a - (d + b)Q$. In this set-up, the RHS is deliberately chosen such that it is $P_{demand} - P_{supply}$ instead of $P_{supply} - P_{demand}$ even though both are the same at equilibrium because both equal to 0. This is done so that the sign convention matches the intended market adjustment direction. When $P_{demand} > P_{supply}$, the net market pressure is positive. When $P_{demand} < P_{supply}$, the net market pressure is negative. This will make the RHS similar to that derived from the Free Fall model and enables the treatment of eqn. 1 as the equilibrium condition of a First-Order Physical System as discussed later.

It is important to note that equation 1 is still only a static equilibrium condition. By itself, it does not describe how the market evolves. The dynamical step occurs when certain "properties" of Free Fall will be imposed on the Market Imbalance equation:

$$\Delta P = P_{demand} - P_{supply}$$

Free Fall: In a Free Fall assuming a linear Drag Force such that $F_d \propto v$ (where v is velocity), Newton's Second Law must be used to model the system. The system has only two forces in action: the Gravitational Force (a constant) and the Linear Drag. A time-displacement differential equation can be used:

$$F_{net} = ma_{net} = \sum F$$

Where m is mass, a_{net} is acceleration, F is force. As mentioned, there are two forces in the system: Gravitational Force F_g and the Drag Force F_d.

$$F_g = mg$$

$$F_d = kv$$

Where g is the Gravitational Acceleration and k is the Proportionality Constant (it is always positive). It must be noted that F_d acts in the direction opposite to the motion of the Free Falling body, thereby slowing it down. So, in the algebraic sum of F_g and F_d, the latter will have a negative sign.

$$ma_{net} = F_g - F_d$$

$$\Rightarrow ma_{net} = mg - kv$$

Dividing by mass m on both sides gives an acceleration equation.

$$\Rightarrow a_{net} = g - \frac{k}{m}v \tag{2}$$

Physically, acceleration is in fact a time derivative of velocity. So:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - \frac{k}{m}v \tag{3}$$

Eqn. 3 is a differential equation with the RHS being $g - \frac{k}{m}v$.

This is where the similarity in the Market expression (eqn. 1) and Free Fall model (eqn. 3) lies: both contain a net imbalance of the form

$$A - Bx$$

where A is a constant driving term and Bx is a linear opposing term. In the Free Fall model, gravity is the driving effect and drag is the opposing effect. In the Market model, Demand and Supply create the corresponding net market pressure.

More importantly, both systems have the qualitative behaviour of a First-Order relaxation system. The net imbalance is positive on one side of equilibrium, negative on the other side, and zero at equilibrium. Therefore, the system is pushed toward the equilibrium point and stops adjusting when that point is reached. This is the deeper resemblance that justifies the Free Fall analogy.

4 Treating the Market as a Free Fall System

Since the RHS of eqn. 1 (Market) and eqn. 3 (Free Fall) are quite similar, eqn. 1 can be treated as a differential equation akin to eqn. 3 by considering both to be analogous to each other, implying certain variable properties of eqn. 3 must be imposed upon eqn. 1.

This imposition is justified by the qualitative behaviour of the idealized Market under *Ceteris Paribus*. Since Demand and Supply are treated as functions of Price/Quantity alone, and since perfect information, instantaneous response, and absence of transaction or information delays are assumed, the Market has no explicit source of delayed feedback, inertia, or oscillation. The market therefore behaves like a First-Order relaxation system: it moves toward equilibrium as the net imbalance decreases.

The Free Fall system under Linear Drag has exactly this kind of behaviour. A Free Falling body accelerates when the gravitational force exceeds drag, slows its acceleration as drag increases, and reaches terminal velocity when the net force becomes zero. Similarly, a Market adjusts when Demand and Supply differ, slows its adjustment as the imbalance decreases, and reaches equilibrium when the net market pressure becomes zero.

Property 1: The differential equation of Free Fall, in essence, is:

$$a_{net}(v) = a_1(v) - a_2(v)$$

Where $a_1(v)$ and $a_2(v)$ are individual acceleration functions of velocity, which are summed according to direction to give net acceleration $a_{net}(v)$. In the Free Fall system, $a_1(v) = g$ (constant) and $a_2(v) \propto v$ (linear drag). However, in the Market analogy, P_{demand} and P_{supply} are both linear functions of Q. It is their algebraic combination that produces an RHS structurally identical to the Free Fall ODE — the analogy operates at the level of the net expression and the resulting First-Order behaviour, not through a literal one-to-one correspondence between individual force terms.

Property 2: The output, a_{net}, is a function of its time integral v. So, essentially, for the purpose of understanding, the differential equation is:

$$\dot{y}(y) = \dot{y}_1(y) - \dot{y}_2(y)$$

A dot over a variable represents a time derivative.

Analogical imposition: Eqn. 1 states that:

$$0 = P_{demand}(Q) - P_{supply}(Q) = c - a - (d + b)Q$$

To convert this static equilibrium condition into a kinematic-like differential equation, it is imposed:

$$\frac{\mathrm{d}Q}{\mathrm{d}\alpha_Q} = P_{net}$$

$$P_{net}(Q) = P_{demand}(Q) - P_{supply}(Q) \tag{4}$$

α_Q is a dummy variable which is introduced to impose the condition. For the Quantity function, it resembles time t (for Price functions, α_P will be used as $\alpha_Q \neq \alpha_P$, which will be proved later). While it isn't exactly t, α_Q (or α_P) can be treated as a parametric variable that defines the "path" of Q (or P) functions and behaves similarly to t in the Free Fall System.

Since this model of P_{net} is linear under *Ceteris Paribus*, it can be directly compared to a_{net}, aiding in proceeding with further calculations.

Analogical Comparison	
Free Fall	Market
a_{net}	P_{net}
$a_1 - a_2$	$P_{demand} - P_{supply}$
v	Q
t	α_Q
g	$c - a$
$\frac{k}{m}$	$d + b$

In mathematics, when two differential equations share identical form, their solution structures are equivalent — meaning any analytical technique valid for one applies directly to the other. Here, the Free Fall equation $\dot{v} = g - \frac{k}{m}v$ and the Market equation $\dot{Q} = (c-a) - (d+b)Q$ are both First-Order linear ODEs with constant coefficients, making them mathematically equivalent as dynamical systems. This is precisely why the Market assumed under *Ceteris Paribus* can be treated as a Physical System.

5 Solving the differential equation

From eqn. 4:

$$\frac{\mathrm{d}Q}{\mathrm{d}\alpha_Q} = c - a - (d + b)Q$$

Let $c - a = \delta$ and $d + b = \varepsilon$.

$$\therefore \frac{\mathrm{d}Q}{\mathrm{d}\alpha_Q} = \delta - Q\varepsilon \tag{5}$$

5.1 Solving via Separable Methods

5.1.1 Solving for $Q(\alpha_Q)$

Eqn. 5 is a simple case of separable variables. Solving eqn. 5 will give Q as a function of the parameter α. Proceeding from eqn. 5:

$$\frac{\mathrm{d}Q}{\delta - Q\varepsilon} = \mathrm{d}\alpha_Q$$

Integrating both sides:

$$\int \frac{\mathrm{d}Q}{\delta - Q\varepsilon} = \int \mathrm{d}\alpha_Q$$

$$-\frac{1}{\varepsilon} \ln|\delta - Q\varepsilon| = \alpha_Q + C_1$$

C_1 is a constant of integration. It depends on the initial conditions. Multiplying both sides by $-\varepsilon$:

$$\ln|\delta - Q\varepsilon| = -\varepsilon\alpha_Q - \varepsilon C_1$$

Taking exponent on both sides:

$$|\delta - Q\varepsilon| = e^{-\varepsilon\alpha_Q - \varepsilon C_1}$$

$$\Rightarrow \delta - Q\varepsilon = Ae^{-\varepsilon\alpha_Q}$$

Where $A = \pm e^{-\varepsilon C_1}$ absorbs the sign from the absolute value.

$$\Rightarrow Q\varepsilon = \delta - Ae^{-\varepsilon\alpha_Q}$$

$$\Rightarrow Q = \frac{\delta}{\varepsilon} - \frac{A}{\varepsilon}e^{-\varepsilon\alpha_Q}$$

Let $\frac{A}{\varepsilon} = D$:

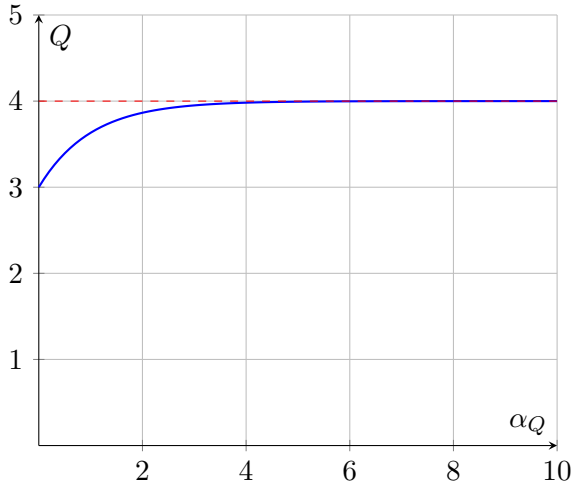
$$\therefore Q(\alpha_Q) = \frac{\delta}{\varepsilon} - De^{-\varepsilon\alpha_Q}$$

To find the equilibrium quantity:

$$Q^* = \lim_{\alpha_Q \to \infty} Q(\alpha_Q) = \frac{\delta}{\varepsilon}$$

This is because, as $\alpha_Q \to \infty$, $De^{-\varepsilon\alpha} \to 0$ as $De^{-\varepsilon\alpha}$ is a strictly decreasing function (in magnitude). The following plot demonstrates this for arbitrarily chosen values of $\frac{\delta}{\varepsilon} = 4$, $D = 1$, and $\varepsilon = 1$.

Plot of $Q(\alpha_Q)$



Although the values are arbitrarily chosen, $\frac{\delta}{\varepsilon} = \frac{c-a}{b+d}$ and D depends on initial conditions. Note that the rate of convergence to equilibrium is governed by $\varepsilon = d + b$: larger values of ε lead to faster stabilization.

Furthermore, it can be observed that regardless of the initial conditions of the differential equation, the Equilibrium Quantity Q^* remains constant. This is exactly what demonstrates the stabilizing behavior of the Market. The initial quantity $Q(0)$ could be anything, but it is guaranteed to lie on P_{supply} and P_{demand} (since both are bijective on $\mathbb{R}^2$), leading to stabilization of the prices due to the "Supplier Force" and the "Consumer Force".

However, with this alone, information regarding the Equilibrium Price P^* is not obtained. But following the exact steps, only with $P^{-1}_{demand}(Q)$ and $P^{-1}_{supply}(Q)$, it is possible to obtain a parametric function of $P(\alpha)$.

5.1.2 Solving for $P(\alpha_P)$

To obtain $P(\alpha_P)$, all that must be done is apply the same steps applied to $P(Q)$ functions, only to their inverse.

$$P^{-1}_{supply}(Q) = Q_{supply}(P) = \frac{P}{b} - \frac{a}{b}$$

$$P^{-1}_{demand}(Q) = Q_{demand}(P) = \frac{c}{d} - \frac{P}{d}$$

Constructing the differential equation:

$$\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = Q_{demand}(P) - Q_{supply}(P)$$

$$\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \frac{c}{d} - \frac{P}{d} - \frac{P}{b} + \frac{a}{b}$$

$$\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \frac{c}{d} + \frac{a}{b} - \left(\frac{1}{d} + \frac{1}{b}\right) P$$

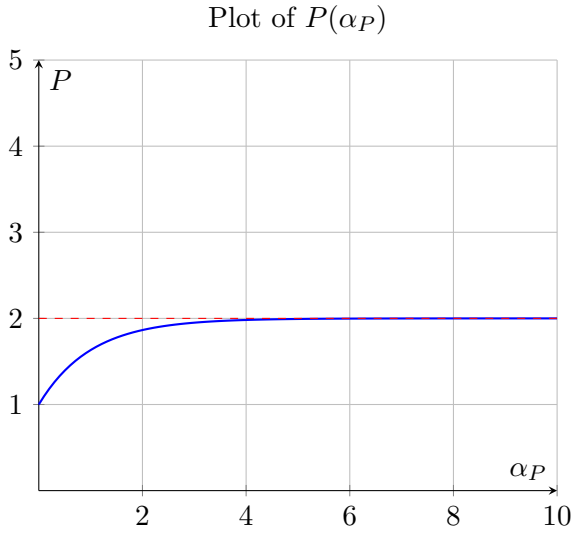
Let $\frac{c}{d} + \frac{a}{b} = \gamma$ and $\frac{1}{d} + \frac{1}{b} = \omega$:

$$\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \gamma - P\omega$$

Solving via the same separable method (with the $\frac{1}{\omega}$ factor from integration):

$$P(\alpha_P) = \frac{\gamma}{\omega} - Ce^{-\omega\alpha_P}$$

For arbitrarily (and perhaps technically inaccurately) chosen values of $\frac{\gamma}{\omega} = 2$, $C = 1$, and $\omega = 1$ for the purpose of demonstrating behaviour:



Akin to the Quantity graph, the Price graph also approaches a stable value of $P^* = \frac{\gamma}{\omega} = 2$ as $\alpha_P \to \infty$.

5.1.3 Proving $\alpha_Q \neq \alpha_P$

Consider that, by solving for $Q(\alpha_Q)$, it is obtained:

$$Q(\alpha_Q) = \frac{\delta}{\varepsilon} - De^{-\varepsilon\alpha_Q}$$

Let the initial condition be $Q(0) = Q_0$.

$$\therefore D = \frac{\delta}{\varepsilon} - Q_0$$

$$\Rightarrow Q(\alpha_Q) = \frac{\delta}{\varepsilon} - (\frac{\delta}{\varepsilon} - Q_0)e^{-\varepsilon\alpha_Q}$$

Now, consider that, by solving for $P(\alpha_P)$, it is obtained:

$$P(\alpha_P) = \frac{\gamma}{\omega} - Ce^{-\omega\alpha_P}$$

The initial condition of P is such that:

$$P(0) = P_0 = \begin{cases} P_{demand}(Q_0) & \text{if } Q_0 < Q^* \quad \text{(shortage)} \\ P_{demand}(Q^*) = P_{supply}(Q^*) = P^* & \text{if } Q_0 = Q^* \quad \text{(at equilibrium)} \\ P_{supply}(Q_0) & \text{if } Q_0 > Q^* \quad \text{(surplus)} \end{cases}$$

These cases rely on the simple fact that: in shortages, the consumer's WTP increases, and the seller accepts the higher prices as the seller's WTA is lesser than "bid" offered by the consumers — the prices are dictated by buyers as there are too many buyers; while during surplus, a rational buyer will not accept a trade at the buyer's price as WTP is less than WTA. Therefore, the seller determines the prices at which the trades would possibly (if at all) occur. For $Q_0 = Q^*$, the Market will already be in equilibrium, so no further developments would occur.

For the following example, it will be assumed that $Q_0 < Q^*$. However, the following proof also applies for surplus conditions.

$$\therefore P_0 = P_{demand}(Q_0) = c - d \cdot Q_0$$

$$\Rightarrow C = \frac{\gamma}{\omega} - (c - d \cdot Q_0)$$

$$\Rightarrow P(\alpha_P) = \frac{\gamma}{\omega} - (\frac{\gamma}{\omega} - c + d \cdot Q_0)e^{-\omega\alpha_P}$$

Since the shortage condition assumed persists till the asymptotic equilibrium:

$$P(\alpha_P) = P_{demand}(Q(\alpha_Q)) \quad \forall \alpha_Q, \alpha_P$$

This also relates to why $P_0 = P_{demand/supply}(Q_0)$.

To show that $\alpha_Q \neq \alpha_P$:

Given that in the shortage regime, the short-side coupling holds for all α:

$$P(\alpha_P) = P_{demand}(Q(\alpha_Q))$$

Substituting the solutions:

$$\underbrace{\frac{\gamma}{\omega} - \left(\frac{\gamma}{\omega} - P_0\right) e^{-\omega\alpha_P}}_{\text{LHS: } P(\alpha_P)} = \underbrace{c - d\left(\frac{\delta}{\varepsilon} - \left(\frac{\delta}{\varepsilon} - Q_0\right) e^{-\varepsilon\alpha_Q}\right)}_{\text{RHS: } P_{demand}(Q(\alpha_Q))}$$

Expanding the RHS:

$$\text{RHS} = c - \frac{d\delta}{\varepsilon} + d\left(\frac{\delta}{\varepsilon} - Q_0\right) e^{-\varepsilon\alpha_Q}$$

Note that $\delta = c - a$ and $\varepsilon = d + b$, so:

$$c - \frac{d\delta}{\varepsilon} = c - \frac{d(c-a)}{d+b} = \frac{c(d+b) - d(c-a)}{d+b} = \frac{cd + cb - dc + da}{d+b} = \frac{cb + da}{d+b}$$

Meanwhile, $\frac{\gamma}{\omega} = \frac{\frac{c}{d} + \frac{a}{b}}{\frac{1}{d} + \frac{1}{b}} = \frac{\frac{cb+da}{db}}{\frac{b+d}{db}} = \frac{cb+da}{b+d}$.

Therefore, the constant terms are equal:

$$c - \frac{d \cdot \delta}{\varepsilon} = \frac{\gamma}{\omega}$$

So the equation reduces to:

$$-\left(\frac{\gamma}{\omega} - P_0\right) e^{-\omega\alpha_P} = d\left(\frac{\delta}{\varepsilon} - Q_0\right) e^{-\varepsilon\alpha_Q}$$

In the shortage regime, $P_0 = P_{demand}(Q_0) = c - dQ_0$, so:

$$\frac{\gamma}{\omega} - P_0 = \frac{\gamma}{\omega} - c + dQ_0 = -\frac{d\delta}{\varepsilon} + dQ_0 = -d\left(\frac{\delta}{\varepsilon} - Q_0\right)$$

Substituting:

$$d\left(\frac{\delta}{\varepsilon} - Q_0\right) e^{-\omega\alpha_P} = d\left(\frac{\delta}{\varepsilon} - Q_0\right) e^{-\varepsilon\alpha_Q}$$

The prefactors cancel (assuming $Q_0 \neq Q^* = \frac{\delta}{\varepsilon}$), leaving:

$$e^{-\omega\alpha_P} = e^{-\varepsilon\alpha_Q}$$

$$\therefore \quad \omega\alpha_P = \varepsilon\alpha_Q$$

Since $\omega = \frac{1}{d} + \frac{1}{b} \neq \varepsilon = d + b$ in general, it follows that:

$$\boxed{\alpha_P \neq \alpha_Q}$$

However, letting $\omega\alpha_P = \varepsilon\alpha_Q = \alpha$, $P(\alpha_P)$ and $Q(\alpha_Q)$ can be rewritten as Price-Alpha and Quantity-Alpha functions such that $P(\alpha_P) = P_\alpha(\alpha)$ and $Q(\alpha_Q) = Q_\alpha(\alpha)$. This enables to plot a Phase Portrait parameterized by α and both the Alpha functions will share the same time-scaling factor β, related to real time by $\alpha = t/\beta$, to provide the Price-Beta $P_\beta(t) = P_\alpha(t/\beta)$ and Quantity-Beta $Q_\beta(t) = Q_\alpha(t/\beta)$ functions.

6 Case examples

To demonstrate how this works, I will present a case example using data Vernon Smith's "Test 1" (*Smith [1962]*) as it is the most convenient paper for this example in that it provides an S&D schedule and also records how the Market evolves with each transaction (which can be treated as a substitute of time). There were no better articles found on the evolution of the Market with time under *Ceteris Paribus*. Therefore, I will also present examples using simulated Continuous Double Auctions (CDAs) which employ a customized algorithm (as discussed in §6.2).

6.1 Experiments with Vernon Smith's data

6.1.1 Methodology

Vernon Smith's "Test 1" provides a chart displaying the S&D schedules as step functions and the price at which goods were traded for the n-th transaction.

S&D step functions can be linearly approximated to obtain P_{demand} and P_{supply} via which the ODE for $P(\alpha_P)$ can be solved. Then, substituting $\omega\alpha_P = \alpha$, the Price-Alpha function $P_\alpha(\alpha)$ can be obtained. Rescaling to real time via $\alpha = t/\beta$ gives the Price-Beta function $P_\beta(t) = P_\alpha(t/\beta)$, which can be regressed against the scatter plot that will be formed by the Price-Transaction chart to empirically obtain the time-scaling factor β, wherein the transaction index is taken to increase linearly with a unit of time t. This treatment is justified by noting that, in a continuous double auction, price evolution occurs over time while transactions arise sequentially as the market adjusts. In the continuous limit, the progression of trades provides a natural ordering of market states, so that transaction index serves as a discrete proxy for temporal evolution. Since the experimental data from Smith's Test 1 records price updates only through the sequence of realized transactions, it is therefore reasonable to parameterize the price path by transaction number. Under this representation, transaction index acts as the discrete analogue of continuous time, with β capturing the effective scaling between transaction steps and the underlying temporal dynamics.

It can be hypothesized that if:

$$\exists\, \beta \in \mathbb{R}^+ \text{ s.t. } P_\beta(t) := P_\alpha\left(\frac{t}{\beta}\right) = P_\alpha(\alpha) \quad \forall\, t \in \mathcal{T}_{\text{CP}},$$

where $\mathcal{T}_{\text{CP}} = \{t \in \mathbb{R}^+ \mid \textit{ceteris paribus} \text{ holds}\}$ and α is the parametric time variable, with the implication that

$$\alpha = \omega\alpha_P = \varepsilon\alpha_Q,$$

then the differential equation

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{1}{\beta\omega}(\gamma - \omega P), \qquad \gamma = \frac{c}{d} + \frac{a}{b}, \;\; \omega = \frac{1}{d} + \frac{1}{b}$$

derived via Newtonian analogy constitutes a valid dynamical model for this Market. The same goes for the inverse:

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{1}{\beta\varepsilon}(\delta - \varepsilon Q), \qquad \delta = c - a, \;\; \varepsilon = d + b$$

6.1.2 Data collection

Referring to Smith [1962], Test 1:

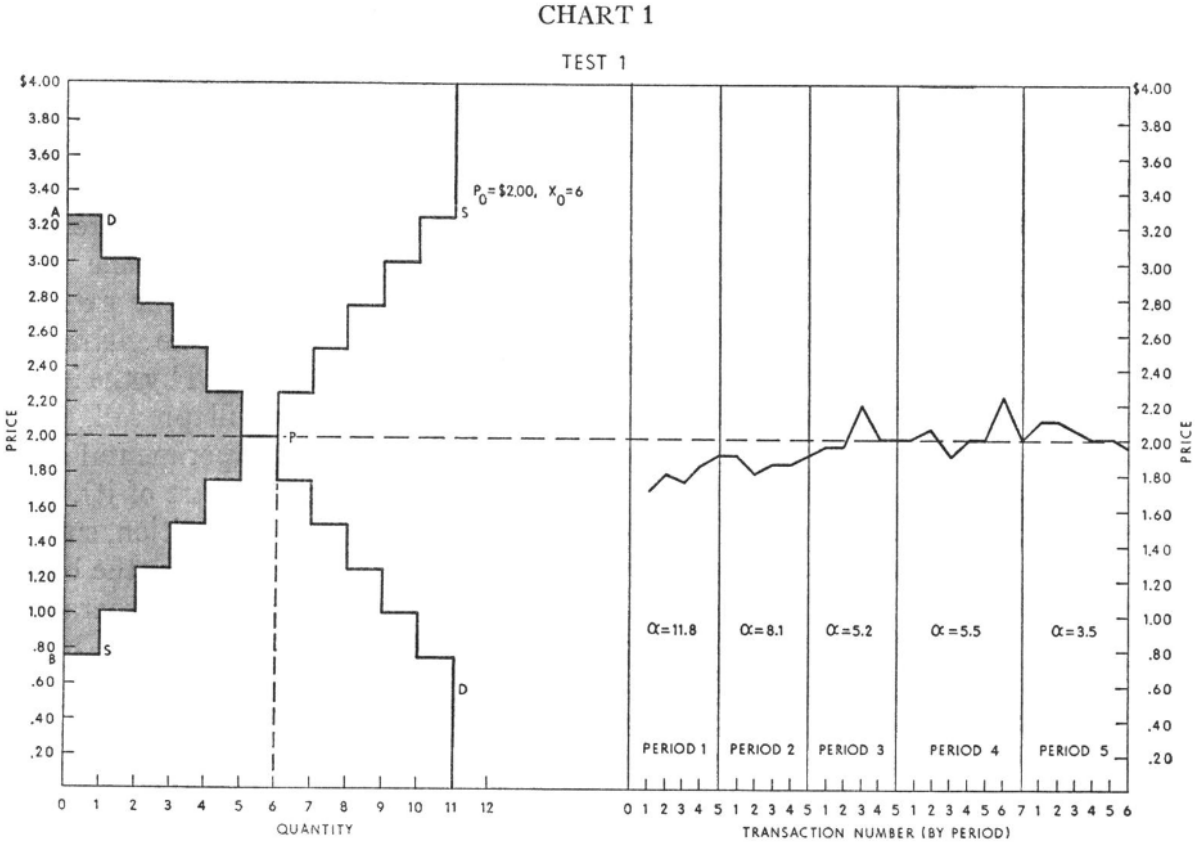


Figure 1: Test 1 chart

This chart of Smith's Test 1 shows the S&D schedule and the Price-Transaction evolution of the Market. It's mentioned and seen that $P^* = 2.00\$$ and $Q^* = 6$.

The S&D schedule can be linearly approximated by reading of points and performing a linear regression as done on Desmos (*Shah [2026a]*):

Q	P_{supply}	P_{demand}
1	0.775	3.225
2	1.000	3.000
3	1.250	2.750
4	1.500	2.467
5	1.767	2.200
6	2.000	2.000
7	2.200	1.767
8	2.467	1.500
9	2.750	1.250
10	3.000	1.000
11	3.225	0.775
	$P_{supply} = 0.2465Q + 0.5149$	$P_{demand} = -0.2465Q + 3.4731$
	$R^2 = 0.9995,\ r = 0.9997$	$R^2 = 0.9995,\ r = -0.9997$

Table 1: Supply and Demand Schedule (symmetric) with Linear Approximations

From the Table 1, the linear approximations of the supply and demand schedules:

$$P_{supply} = a + bQ, \qquad a = 0.5149, \quad b = 0.2465$$
$$P_{demand} = c - dQ, \qquad c = 3.4731, \quad d = 0.2465$$

From Figure 1, the Price-Transaction scatter plot is obtained by reading off values directly. Two modifications are made to the original transaction numbering: trading periods are disregarded and transactions are indexed serially across the entire experiment, and each index is decremented by 1 such that $t = 0$ corresponds to the market's initial state — consistent with the continuous-time interpretation established above. The data table and the scatter plot created by reading off values are:

Transaction	t	P (\$)
1	0	1.700
2	1	1.800
3	2	1.750
4	3	1.850
5	4	1.900
6	5	1.880
7	6	1.800
8	7	1.860
9	8	1.860
10	9	1.900
11	10	1.940
12	11	1.940
13	12	2.200
14	13	2.000
15	14	2.000
16	15	2.000
17	16	2.060
18	17	1.900
19	18	2.000
20	19	2.000
21	20	2.225
22	21	2.000
23	22	2.100
24	23	2.100
25	24	2.095
26	25	2.000
27	26	2.000
28	27	1.950

Table 2: Price-Transaction Data

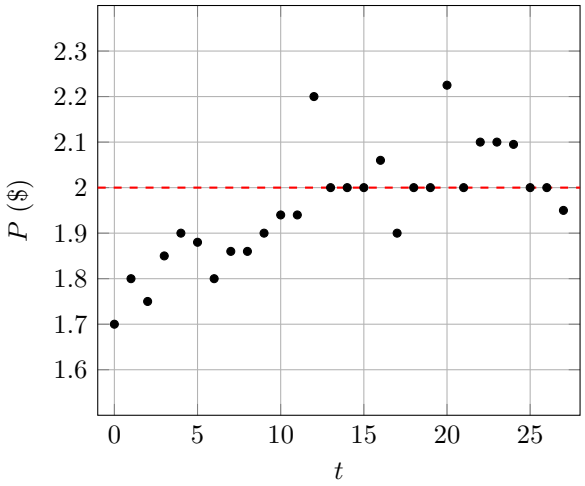


Figure 2: Price-Transaction Scatter Plot

The experimental results of Smith's Price evolution are seen as the scatter plot in Table 3. This is what the ODE model will be tested against for this experiment according to the hypothesis to find the time-scalar β.

6.1.3 Solution via differential equation

From the data collected, the derived parameters ("Greeks") are:

$$\gamma = \frac{c}{d} + \frac{a}{b} = \frac{3.4731}{0.2465} + \frac{0.5149}{0.2465} = 16.180$$
$$\omega = \frac{1}{d} + \frac{1}{b} = \frac{1}{0.2465} + \frac{1}{0.2465} = 8.114$$
$$\delta = c - a = 3.4731 - 0.5149 = 2.9582$$
$$\varepsilon = d + b = 0.2465 + 0.2465 = 0.4930$$

As a consistency check, the equilibrium price and quantity are recovered exactly:

$$P^* = \frac{\gamma}{\omega} = \frac{16.180}{8.114} \approx 2.000\checkmark$$
$$Q^* = \frac{\delta}{\varepsilon} = \frac{2.9582}{0.4930} \approx 6.000\checkmark$$

These Greeks can be used to construct and derive Price functions: $P(\alpha_P)$, $P_\alpha(\alpha)$, $P_\beta(t) = P_\alpha(t/\beta)$.

Having solved the ODE in section 5.1, the Greeks' values can be directly substituted to first construct $P(\alpha_P)$:

$$\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \gamma - \omega P$$

has the general solution

$$P(\alpha_P) = \frac{\gamma}{\omega} + Ce^{-\omega\alpha_P}$$

Substituting $\gamma = 16.180$ and $\omega = 8.114$ exactly:

$$\frac{\gamma}{\omega} = \frac{16.180}{8.114} = 1.9941\ldots \approx 2.000$$

where the approximation error is $\epsilon_1 = |2.000 - 1.9941| \approx 5.9 \times 10^{-3}$, which is negligibly small. Applying the initial condition $P(0) = 1.700$:

$$1.700 = \frac{16.180}{8.114} + C$$
$$C = 1.700 - \frac{16.180}{8.114} = -0.2941\ldots \approx -0.300$$

where the approximation error is $\epsilon_2 = |-0.300 - (-0.2941)| \approx 5.9 \times 10^{-3}$, again negligibly small. The solution is therefore

$$P(\alpha_P) = \frac{16.180}{8.114} - 0.2941\, e^{-8.114\,\alpha_P}$$

or in approximate closed form:

$$P(\alpha_P) \approx 2.000 - 0.300\, e^{-8.114\,\alpha_P}$$

The errors arise from the approximation of S&D schedules and "reading off" values from the charts, but both approximation errors are on the order of 10^{-3}, arising solely from rounding in the regression-derived parameters a, b, c, d, and are negligible for the purposes of empirical validation. From this, the Price-Alpha function $P_\alpha(\alpha)$ can be derived:

$$P_\alpha(\alpha) \approx 2.000 - 0.300e^{-\alpha}$$

The Price-Alpha function can now be regressed against the scatter plot in Table 3 to find the β of the Price-Beta function $P_\beta(t) = P_\alpha(t/\beta)$ as done on Desmos Shah [2026a]:

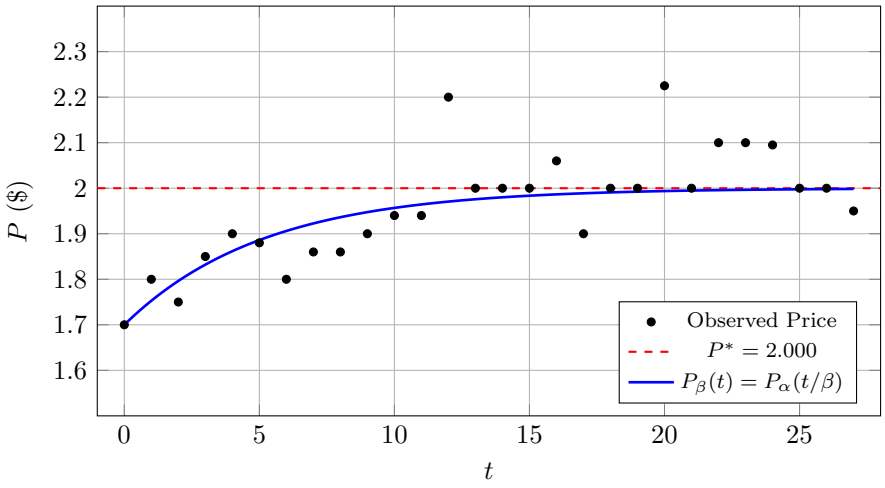


Figure 3: Price-Transaction scatter plot with fitted $P_\beta(t)$

Regressing $P_\beta(t) = P_\alpha(t/\beta) = 2 - 0.3000\, e^{-t/\beta}$ against the Price-Transaction scatter plot yields $\beta = 5.1658$ with $R^2 = 0.5614$. The modest R^2 is expected given the noise inherent in a single experimental trading period; the curve nonetheless captures the convergence trend toward $P^* = 2.000$ and also captures the moment around $t = 5$ when the Price curve "bends" towards convergence.

A percent-error plot can be created:

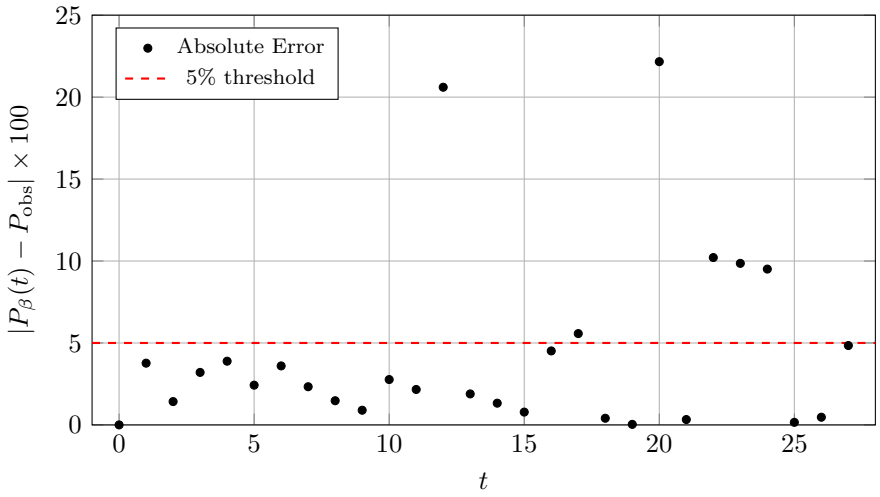


Figure 4: Absolute percentage error $|P_\beta(t) - P_{\text{obs}}| \times 100$ per transaction

As seen from Figure 4, most of the prediction lies within a 5% error range. The points above the error margin can be attributed to random noise and also "reading off" errors. However, as stated, the ODE curve aptly captures the motion and the path of the Market to convergence from a point in the shortage regime.

6.1.4 Phase Portrait

A Phase Portrait is crucial in Dynamical Systems. It is used to plot velocity vs. acceleration, displacement vs. velocity, etc. However, for our Dynamic System of this Economic Market, we can create a Phase Portrait of Price vs. Quantity.

The Quantity-Alpha function $Q_\alpha(\alpha)$ must be obtained to plot the Market Phase Portrait curve $\vec{M}(\alpha) =< Q_\alpha(\alpha), P_\alpha(\alpha) >$. To find the Quantity functions, the initial condition of Quantity $Q(0)$ must be found.

Upon observation, it can be seen that no explicit mention of initial traded Quantity is present. However, the initial condition of Q is similarly determined by the short-side rule:

$$Q(0) = Q_0 = \begin{cases} Q_{supply}(P_0) & \text{if } P_0 < P^* \quad \text{(shortage)} \\ Q_{supply}(P^*) = Q_{demand}(P^*) = Q^* & \text{if } P_0 = P^* \quad \text{(at equilibrium)} \\ Q_{demand}(P_0) & \text{if } P_0 > P^* \quad \text{(surplus)} \end{cases}$$

By the same reasoning as in 5.1.3 to determine $P(0)$, in a shortage ($P_0 < P^*$), supply is the short side and constrains realized quantity; in a surplus ($P_0 > P^*$), demand is the short side. At $P_0 = P^*$, both schedules agree and the market is at full equilibrium.

From Test 1, since $P_0 < P^*$, the Market starts off from the shortage regime.

$$\therefore Q_0 = Q_{supply}(P_0) = -\frac{a}{b} + \frac{P_0}{b}$$

$$\Rightarrow Q_0 = -\frac{0.5149}{0.2465} + \frac{1.7}{0.2465} \approx 4.807$$

Having found Q_0, $Q(\alpha_P)$ and then $Q_\alpha(\alpha)$ can be constructed with the Latins (a, b, c, d) and the Greeks. However, another approach can be used in which $P_\alpha(\alpha)$ can be substituted in the appropriate linear $Q(P)$ schedule based on the regime being a shortage or surplus to directly find $Q_\alpha(\alpha)$. As mentioned in 5.1.3, this works because the Market will belong to a singular regime all the way up to equilibrium. Therefore, Quantity will completely follow the apt $Q(P)$ schedule as determined by the particular regime all the way up to equilibrium. Since the Market in Test 1 is in the shortage regime:

$$Q_\alpha(\alpha) = Q_{supply}(P_\alpha(\alpha))$$

$$\Rightarrow Q_\alpha(\alpha) = -\frac{a}{b} + \frac{P_\alpha(\alpha)}{b}$$

$$\Rightarrow Q_\alpha(\alpha) \approx 6.024 - 1.217e^{-\alpha}$$

It can be noted that $\lim_{\alpha\to\infty} Q_\alpha(\alpha) \approx Q^* = 6$. The $Q_\beta(t) = Q_\alpha(t/\beta)$ plot is:

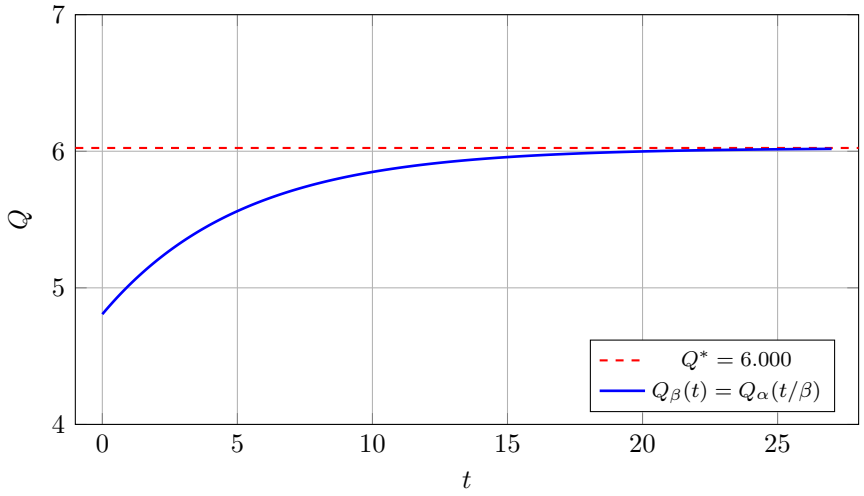


Figure 5: Theoretical quantity evolution $Q_\beta(t)$. No experimental quantity data is available from Smith's Test 1, as only transaction prices are recorded. The scaling factor $\beta = 5.1658$ is adopted directly from the Price-Alpha regression, since P_α and Q_α share a common parametric variable α and are therefore governed by the same time-scaling.

The α-parameterized Market Phase Portrait curve $\vec{M}(\alpha) =< Q_\alpha(\alpha), P_\alpha(\alpha) >$ can be plotted as:

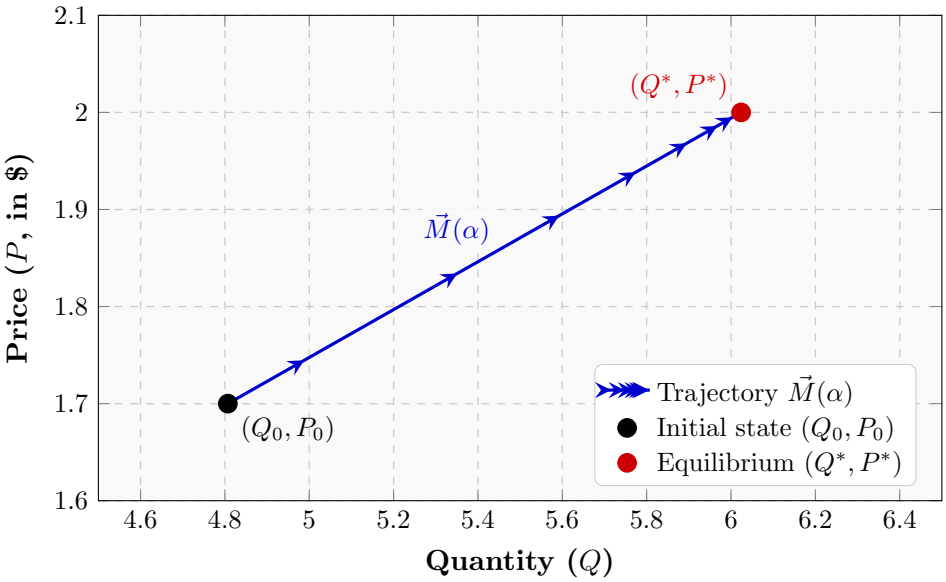


Figure 6: Market phase portrait $\vec{M}(\alpha) = \langle Q_\alpha(\alpha), P_\alpha(\alpha) \rangle$

The Phase Portrait $\vec{M}(\alpha)$ is a linear curve starting from (Q_0, P_0) and ending at (Q^*, P^*). Linearity is ensured as the S&D schedules are linear under *Ceteris Paribus*. Furthermore, this linearity also coincides with the fact that the velocity vs. acceleration Phase Plots for Free-Fall with linear drag are also linear in shape, highlighting the similarities between a Market under *Ceteris Paribus* and a body in Free-Fall.

6.2 Computerized experiments using customized CDA

As much as I, as the author, would have liked to include more experiments of Vernon Smith from *Smith [1962]*, I found it particularly difficult to extract data for all experiments after "Test 1". Therefore, while keeping the spirit of Double Auction alive, this paper employs a computerized CDA to record the evolution of Price curves.

6.2.1 Methodology

The purpose of the computerized Continuous Double Auction (CDA) experiment is to test whether the First-Order convergence structure derived earlier continues to appear when the market is not centrally cleared, but instead evolves through decentralized sequential trading mediated by a public order book. In the deterministic development, the market was treated as an idealized First-Order relaxation system under *Ceteris Paribus*, perfect information, and absence of external shocks or delays. However, an actual double auction does not update price as a smooth function. It evolves through discrete bids, asks, acceptances, and transaction prices. Therefore, the computerized CDA is used as a bridge between the ideal ODE model and a more realistic sequential market mechanism.

The key methodological idea is the following: the ODE model supplies the theoretical convergence envelope, while the CDA supplies a stochastic realized price path. The simulated market is then evaluated by asking whether the realized CDA prices converge toward the same equilibrium price P^*, whether an exponential envelope can describe the post-formation trend, and whether the deviations from that envelope behave like practical market noise rather than a systematic contradiction of the model.

Supply and demand environment. The computerized market uses the same linear supply and demand framework developed earlier in the paper:

$$P_{supply}(Q) = a + bQ, \qquad P_{demand}(Q) = c - dQ,$$

where

$$a, b, c, d \in \mathbb{R}^+.$$

The competitive equilibrium is obtained by solving

$$P_{supply}(Q^*) = P_{demand}(Q^*).$$

Therefore,

$$a + bQ^* = c - dQ^*$$

and hence

$$Q^* = \frac{c - a}{b + d}.$$

The equilibrium price is

$$P^* = a + bQ^* = c - dQ^*,$$

which may also be written as

$$P^* = \frac{bc + ad}{b + d}.$$

The simulation creates N buyers and N sellers. Buyer values and seller costs are generated from the demand and supply curves using midpoint indexing:

$$v_i = c - d(i + 0.5), \qquad i = 0, 1, \ldots, N - 1,$$

and

$$\kappa_j = a + b(j + 0.5), \qquad j = 0, 1, \ldots, N - 1.$$

Here, v_i is buyer i's private value and κ_j is seller j's private cost. A buyer is willing to purchase only at prices

$$p \leq v_i,$$

and a seller is willing to sell only at prices

$$p \geq \kappa_j.$$

Thus, buyer surplus and seller surplus are

$$\pi_i^B = v_i - p,$$

and

$$\pi_j^S = p - \kappa_j.$$

A trade is individually rational when

$$\pi_i^B \geq 0, \qquad \pi_j^S \geq 0.$$

For the equilibrium comparison to be meaningful inside the finite simulated market, the population size is chosen such that

$$0 < Q^* < N.$$

If $Q^* > N$, the competitive equilibrium lies outside the generated population, and convergence to P^* is not a valid test of the simulated CDA because the finite market does not contain enough traders to support the theoretical equilibrium quantity.

Trader spacing convention: The midpoint indexing $v_i = c - d(i+0.5)$ and $\kappa_j = a + b(j+0.5)$ generates one trader per unit of quantity on each side: trader j on the supply side occupies the position $Q = j + 0.5$ along the supply schedule, so consecutive sellers are spaced by $\aleph = 1$ goods per trader. This convention makes trader-index offsets and quantity offsets numerically interchangeable throughout the paper, so that, for example, the cost of the k-th extramarginal seller is $\kappa_{j^*+k} = P_{supply}(Q^* + k\,\aleph) = P_{supply}(Q^* + k)$ with $\aleph = 1$. Any other choice $\aleph \neq 1$ (e.g. a sparser or denser trader population for the same schedule) would require carrying the $\aleph$ factor explicitly in offset expressions of §6.2.2.

Core CDA institution. The computerized market is implemented as a corrected sequential CDA. The order book begins empty. At each event, one active trader is randomly selected. With probability 1/2 the selected trader is a buyer; with probability 1/2 the selected trader is a seller, conditional on there being active traders on the relevant side of the market.

The book state at event t is represented by the best bid and best ask:

$$B_t^* = \max\{B_{i,t}\},$$

$$A_t^* = \min\{A_{j,t}\}.$$

If no bid has been posted, B_t^* is undefined. If no ask has been posted, A_t^* is undefined.

The CDA execution rules are:

1. A buyer may accept the current best ask if it is individually rational.
2. A seller may accept the current best bid if it is individually rational.
3. If a buyer submits a bid that crosses the best ask, the trade executes at the standing ask.
4. If a seller submits an ask that crosses the best bid, the trade executes at the standing bid.
5. After a trade, both matched agents exit the market.
6. A crossed order book is not allowed to persist.

The third and fourth rules are important. In a CDA, the realized transaction price is the standing quote that gets accepted, not a Walrasian clearing price. Therefore, if a buyer submits a bid satisfying

$$B_{i,t}^{new} \geq A_t^*,$$

then the trade price is

$$p_t = A_t^*.$$

Similarly, if a seller submits an ask satisfying

$$A_{j,t}^{new} \leq B_t^*,$$

then the trade price is

$$p_t = B_t^*.$$

This ensures that the simulation remains a CDA rather than a centralized batch-clearing mechanism.

Behavioral layer: frustration and aggressiveness. In addition to the CDA institution, the simulation introduces a behavioral state variable designed to mimic human impatience, frustration, and increasing willingness to act after unsuccessful attempts. For each buyer and seller, an aggressiveness parameter is maintained:

$$s_i^B \geq 1, \qquad s_j^S \geq 1.$$

This is not a standard textbook CDA assumption. It is a constructed behavioral proxy intended to represent the idea that a trader who cannot execute or improve the market may become more aggressive in future actions.

The behavioral state affects both quote improvement and acceptance probability. A higher value of s_i makes a buyer or seller more aggressive, but the state is bounded above:

$$s_i \leq s_{\max}.$$

In the implemented simulations,

$$s_{\max} = 4.$$

When a trader attempts to act but cannot improve the book, the trader's behavioral state is updated by

$$s_i \leftarrow \min\left(s_{\max},\ s_i \cdot 1.5^{1/5}\right).$$

This is a simplified event-based version of the earlier rejection-counter idea:

$$s(n+1) = s(n) \cdot 1.5^{k/5},$$

where k represents the number of failed or rejected attempts. The event-based rule is used because the CDA is not a fixed bilateral matching process. A trader may fail to improve the book not because a specific counterparty rejected them, but because the live book already contains a better quote. The update rule therefore treats failure to improve the quote as the CDA analogue of a failed attempt.

This behavioral construction is deliberately bounded. Its purpose is not to make agents irrational. Rather, it gives agents a controlled adaptation mechanism:

$$\text{failed attempt} \quad \Rightarrow \quad \text{higher future aggressiveness.}$$

If a trader successfully trades, they leave the market and their state no longer evolves.

Quote-posting rules. If a buyer acts and no standing bid exists, the buyer's first bid is initialized as a fraction of their private value:

$$B_{i,t}^{new} = \phi_B v_i,$$

where

$$0 < \phi_B \leq 1.$$

If a standing bid already exists, the buyer attempts to improve the best bid according to

$$B_{i,t}^{new} = B_t^* + \delta_0 s_i^B,$$

where $\delta_0 > 0$ is the base quote tick. The bid is capped by the buyer's value:

$$B_{i,t}^{new} = \min\left(B_{i,t}^{new}, v_i\right).$$

Similarly, if a seller acts and no standing ask exists, the seller's first ask is initialized as a multiple of their private cost:

$$A_{j,t}^{new} = \phi_S \kappa_j,$$

where

$$\phi_S \geq 1.$$

If a standing ask already exists, the seller attempts to improve the best ask by lowering it:

$$A_{j,t}^{new} = A_t^* - \delta_0 s_j^S.$$

The ask is floored by the seller's cost:

$$A_{j,t}^{new} = \max\left(A_{j,t}^{new}, \kappa_j\right).$$

The initial-condition parameters ϕ_B and ϕ_S are distinct from the parametric Free-Fall variable α used in earlier sections; they are dimensionless quote-initialization multipliers specific to the CDA experiment. A low ϕ_B produces conservative initial bids, while a high ϕ_S produces high initial asks. The same supply and demand curves can therefore be tested under multiple starting conditions.

Acceptance probability functions. A trader does not automatically accept every profitable quote. Instead, acceptance is probabilistic and depends on the trader's surplus opportunity and behavioral state.

If a buyer observes a best ask A_t^* satisfying

$$A_t^* \leq v_i,$$

then the buyer accepts the ask with probability

$$\Pr(\text{buyer } i \text{ accepts } A_t^*) = \min\left(1,\ s_i^B \max\left(0, 1 - \frac{A_t^*}{v_i}\right)\right).$$

This probability increases when the ask is far below the buyer's value and also increases when the buyer's aggressiveness state s_i^B is high.

If a seller observes a best bid B_t^* satisfying

$$B_t^* \geq \kappa_j,$$

then the seller accepts the bid with probability

$$\Pr(\text{seller } j \text{ accepts } B_t^*) = \min\left(1,\ s_j^S \max\left(0, 1 - \frac{\kappa_j}{B_t^*}\right)\right).$$

This probability increases when the bid is far above the seller's cost and when the seller's aggressiveness state s_j^S is high. Both acceptance functions are bounded in the interval $[0, 1]$, so they are valid probabilities:

$$0 \leq \Pr(\text{acceptance}) \leq 1.$$

Other monotonic acceptance functions can also be constructed which satisfy this property; however, those functions are not explored in this paper as monotonicity of acceptance functions would guarantee similar behavior of agents as compared among various monotonic acceptance functions.

Algorithmic specification. A single simulation run proceeds as follows:

1. Choose linear supply and demand parameters a, b, c, d.
2. Compute P^* and Q^*.
3. Generate buyer values $\{v_i\}_{i=0}^{N-1}$ and seller costs $\{\kappa_j\}_{j=0}^{N-1}$.
4. Initialize all buyers and sellers as active.
5. Initialize all behavioral states:
$$s_i^B = 1, \qquad s_j^S = 1.$$
6. Initialize the order book as empty.
7. At each event, randomly select an active buyer or seller.
8. If a buyer is selected:
   (a) Check whether the best ask can be accepted.
   (b) If accepted, execute at A_t^*.
   (c) If not accepted, compute a new bid using the quote-posting rule.
   (d) If the new bid crosses the best ask, execute at A_t^*.
   (e) Otherwise, post the bid if it improves the book.
   (f) If the bid does not improve the book, update s_i^B.
9. If a seller is selected:
   (a) Check whether the best bid can be accepted.
   (b) If accepted, execute at B_t^*.
   (c) If not accepted, compute a new ask using the quote-posting rule.
   (d) If the new ask crosses the best bid, execute at B_t^*.
   (e) Otherwise, post the ask if it improves the book.
   (f) If the ask does not improve the book, update s_j^S.
10. Record each realized transaction price p_t.
11. Stop when the maximum event count is reached, one side of the market is exhausted, or trading becomes inactive after most feasible equilibrium volume has traded.

This algorithm is reproducible because every random event is generated from a fixed pseudo-random seed. For a given seed, market parameters, initial-condition parameters, and simulation settings, the sequence of trades is deterministic and can be exactly regenerated.

No leakage of theoretical equilibrium. A crucial condition of the simulation is that agents do not use the theoretical equilibrium price. The variables P^* and Q^* are computed for post-simulation evaluation only. They do not enter the trader decision rules.

Agents observe only:
$$v_i, \quad \kappa_j, \quad B_t^*, \quad A_t^*, \quad s_i, \quad \delta_0,$$
and random activation. Agents do not observe:
$$P^*, \quad Q^*, \quad \beta, \quad \text{future prices}, \quad \text{ODE residuals}.$$

This matters because the purpose of the CDA experiment is to see whether convergence to P^* emerges from decentralized trading rather than being imposed by the model. Even the behavioral state s_i updates only from local trading conditions. It is not adjusted according to distance from P^*.

Order-book formation. The early part of a CDA simulation is a book-formation stage. At the beginning of the run, there may be no meaningful best bid or best ask, and observed transaction prices may be dominated by initial quote placement rather than market convergence. For this reason, the ODE is not fitted immediately from the first transaction.

The order book is treated as formed once the live spread becomes persistently small. The live spread is defined as

$$\text{Spread}_t = A_t^* - B_t^*.$$

The simulation samples this spread over event time and computes a rolling median. Book formation is detected when

$$\text{Median}\left(\text{Spread}_{t:t+w}\right) < \eta,$$

where w is the rolling window and η is the spread threshold. A minimum burn-in condition is also imposed:

$$n \geq n_{\min},$$

where n is the transaction index. This prevents the model from fitting the ODE too early. This formation criterion uses only order-book information. It does not use P^*, and therefore does not introduce equilibrium leakage.

Post-formation ODE fit. After book formation, the realized transaction-price series is smoothed using a moving average. If p_n is the n-th realized CDA transaction price, then the moving average is

$$\bar{P}_n = \frac{1}{m} \sum_{\ell=0}^{m-1} p_{n-\ell},$$

where m is the moving-average window. The theoretical post-formation convergence envelope is

$$P_\beta(t) = P_\alpha(t/\beta) = P^* + (P_0 - P^*)\, e^{-(t-n_0)/\beta}, \qquad t = n.$$

Here,

$$P_0 = \bar{P}_{n_0},$$

where n_0 is the detected post-formation starting index. The only fitted parameter is

$$\beta > 0.$$

The parameter β is the empirical time-scaling factor between transaction index and the underlying continuous convergence process. The fitted value of β is chosen to minimize the sum of squared residuals:

$$\hat{\beta} = \arg\min_{\beta>0} \sum_{n=n_0}^{N_T} \left[\bar{P}_n - \left(P^* + (P_0 - P^*)\, e^{-(n-n_0)/\beta}\right)\right]^2.$$

This is a one-parameter nonlinear least-squares regression.

Residual definition and statistics: The residual at transaction index n is defined as

$$e_n = \bar{P}_n - P_\beta(n).$$

Residuals measure the difference between the realized stochastic CDA path and the ideal deterministic convergence envelope. Because the CDA is sequential, stochastic, and finite-agent, residuals are expected. They are not by themselves a failure of the model. The key question is whether the residuals are small in price units and whether positive and negative residual regions approximately balance.

The first evaluation statistic is the final deviation from equilibrium:

$$\Delta P_{\text{final}} = \frac{1}{L} \sum_{n=N_T-L+1}^{N_T} p_n - P^*,$$

where L is the final averaging window. The mean squared error and root mean squared error are

$$\text{MSE} = \frac{1}{M} \sum_{n=n_0}^{N_T} e_n^2, \qquad \text{RMSE} = \sqrt{\text{MSE}}.$$

The signed residual integral and absolute residual area, approximated by the trapezoidal rule, are

$$I_{\text{signed}} \approx \sum_{r=1}^{M-1} \frac{e_r + e_{r-1}}{2} (n_r - n_{r-1}), \qquad I_{\text{abs}} \approx \sum_{r=1}^{M-1} \frac{|e_r| + |e_{r-1}|}{2} (n_r - n_{r-1}).$$

A useful residual-balance statistic is therefore

$$\rho_{\text{bal}} = \frac{|I_{\text{signed}}|}{I_{\text{abs}}}.$$

When $\rho_{\text{bal}} \approx 0$, positive and negative residual regions are approximately balanced. When $\rho_{\text{bal}} \approx 1$, the residuals are mostly one-sided, suggesting systematic bias. The coefficient of determination R^2 is also reported, but is not the primary success metric: a low or negative R^2 is expected in some runs because the realized CDA path is stochastic, while the fitted ODE has only one free parameter. Convergence toward P^*, RMSE, and residual balance are more informative.

Monte Carlo evaluation: Because the CDA contains random activation and probabilistic acceptance, a single run is not sufficient to characterize the market. For each supply-demand environment and initial condition, the simulation is therefore repeated across R random seeds:

$$r = 1, 2, \ldots, R.$$

For each run, the statistics $\hat{\beta}^{(r)}, \Delta P_{\text{final}}^{(r)}, \text{RMSE}^{(r)}, I_{\text{signed}}^{(r)}, I_{\text{abs}}^{(r)}$ are recorded, and Monte Carlo means and standard deviations are reported.

Assumptions supporting the CDA experiment.

1. ***Ceteris Paribus.*** The supply and demand curves are fixed during a simulation.
2. **Linear Schedules.** Supply and demand are linear over the relevant admissible domain.
3. **Finite but Large Market.** Q^* lies inside the population.
4. **Private Values and Costs.** Each buyer's value and seller's cost is fixed.
5. **Individual Rationality.** Buyers do not buy above value; sellers do not sell below cost.

6. **No Equilibrium Leakage.** Agents do not observe or use P^*, Q^*, or the ODE fit.
7. **Sequential CDA Institution.** Prices are formed through bids, asks, and accepted standing quotes, not Walrasian clearing.
8. **Random Activation.** The order in which traders act is stochastic.
9. **Behavioral Adaptation.** The state variable s_i approximates bounded human frustration.
10. **Post-Simulation Evaluation.** The ODE is fitted only after trades have occurred.

6.2.2 Monte Carlo CDA Testing: Five Supply–Demand Environments

To test whether the computerized CDA behavior is robust to random activation and probabilistic acceptance, 750 Monte Carlo simulations were run: 5 supply–demand environments × 2 initial-condition regimes (shortage, surplus) × 75 random seeds. The two initial-condition regimes correspond to

$$(\phi_B, \phi_S)_{\text{shortage}} = (0.50, 1.05), \qquad (\phi_B, \phi_S)_{\text{surplus}} = (0.95, 1.50).$$

These parameters do not change the underlying supply and demand curves. They only change the initial quote behavior of agents in the CDA. All 750 runs passed the order-book audit with zero crossed-book violations. The complete simulation pipeline, including the audited CDA mechanism, the formation-detection routine, and the post-formation β-regression, is available at Shah [2026b].

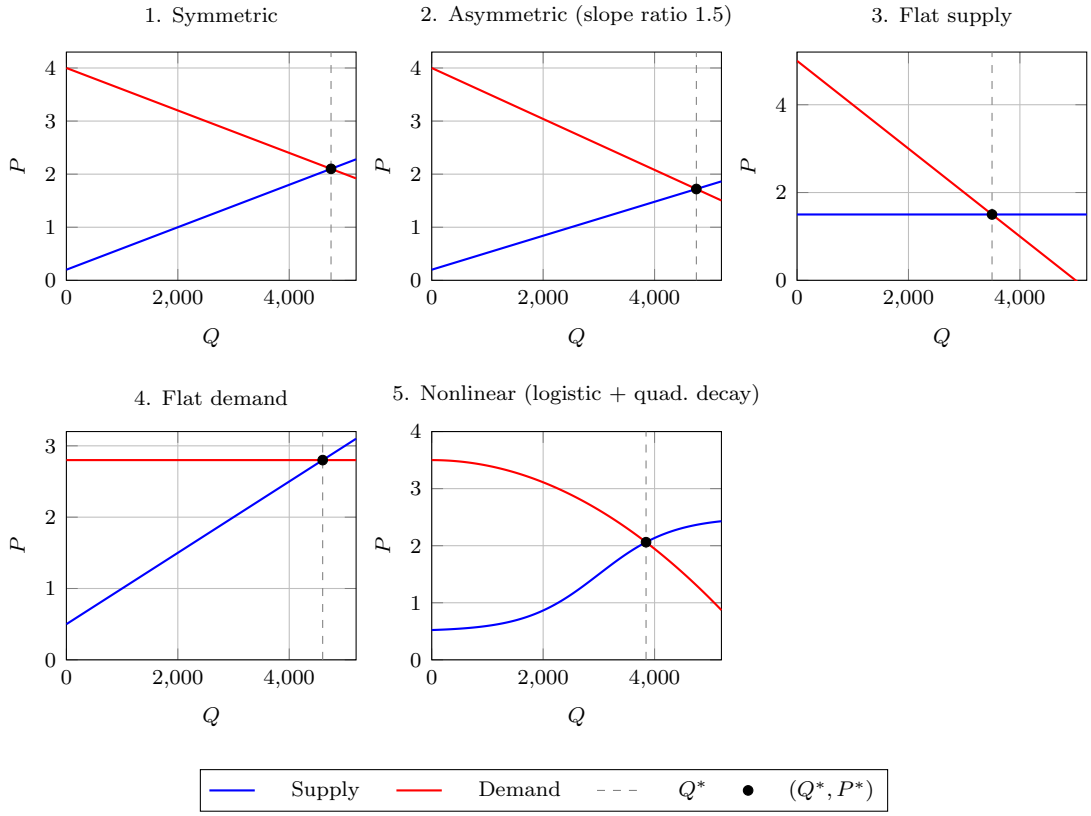


Figure 7: The five supply–demand environments used in the CDA Monte Carlo experiment. Black dot in each panel marks the schedule intersection (Q^*, P^*).

Environment 1 — Symmetric S&D The parameters are $a = 0.20$, $b = 0.00040$, $c = 4.00$, $d = 0.00040$. The slopes are equal in magnitude, so the supply and demand curves are reflections of each other through the line $P = P^*$; this is the most symmetric and theoretically cleanest case.

$$\gamma = \tfrac{c}{d}+\tfrac{a}{b} = 10500.00, \qquad \omega = \tfrac{1}{d}+\tfrac{1}{b} = 5000.00, \qquad \delta = c-a = 3.80, \qquad \varepsilon = d+b = 8.00\times10^{-4}.$$

The equilibrium price and quantity are recovered as $P^* = \gamma/\omega = 2.1000$ and $Q^* = \delta/\varepsilon = 4750$.

The analytical Price-Alpha solution to $\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \gamma - \omega P$, written with $\alpha = \omega\alpha_P$, has the closed form

$$P_\alpha(\alpha) = P^* - (P^* - P_0)\, e^{-\alpha} = 2.1000 - (2.1000 - P_0)\, e^{-\alpha}.$$

The order-book formation diagnostic (mean live spread across 75 seeds) is shown in Figure 8. The rolling-median spread drops to a small persistent value within the first ∼1000 events, corresponding to a contract transaction index of $n_0 = 250$ used throughout the post-formation regression.

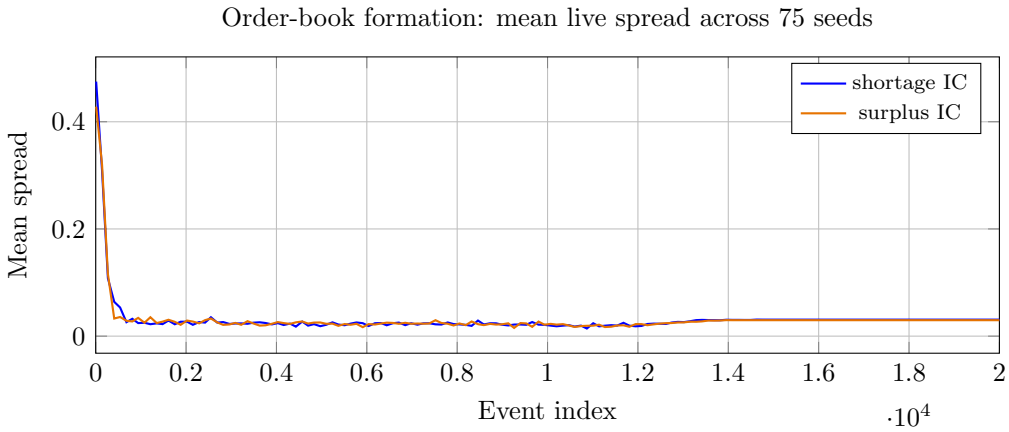


Figure 8: Order-book formation diagnostic for the symmetric linear market.

Regressing β on the 75-seed mean post-formation path yields:

$$\text{Shortage IC:}\quad P_\beta(t) = P_\alpha(t/\beta) = 2.1000 - (2.1000 - 1.9312)\, e^{-(t-n_0)/\beta}, \quad \beta = 149.70,\ n_0 = 250,$$
$$\text{Surplus IC:}\quad P_\beta(t) = P_\alpha(t/\beta) = 2.1000 - (2.1000 - 2.2821)\, e^{-(t-n_0)/\beta}, \quad \beta = 162.34,\ n_0 = 250.$$

Figure 9 shows the 75 individual seed paths (faint), the within-regime averages (bold), and the regressed ODE fits (dashed), with shortage and surplus on the same axes. Figure 10 shows the per-seed residuals against the mean-path ODE fit.

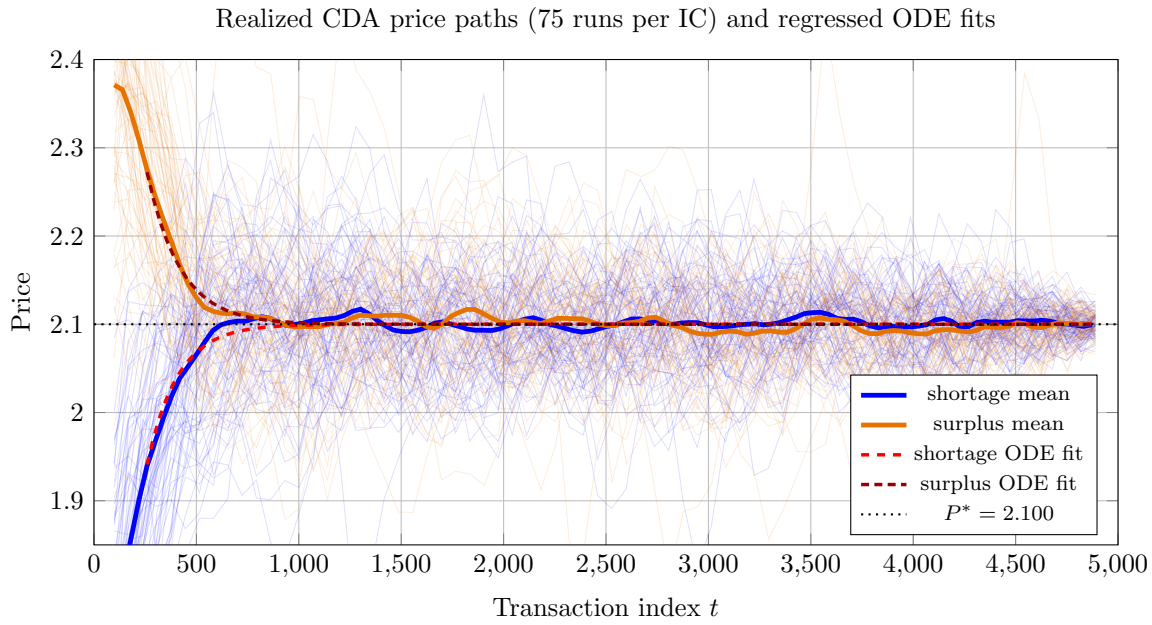


Figure 9: Symmetric linear S&D: 75 individual 100-transaction moving-average price paths per initial-condition regime (faint), with the within-regime means (bold solid) and regressed ODE fits $P_\beta(t)$ (dashed). Shortage IC: $\beta = 149.70$, $R^2 = 0.93$, RMSE $= 0.0064$. Surplus IC: $\beta = 162.34$, $R^2 = 0.93$, RMSE $= 0.0069$. With symmetric slopes, the realised mean tracks P^* throughout.

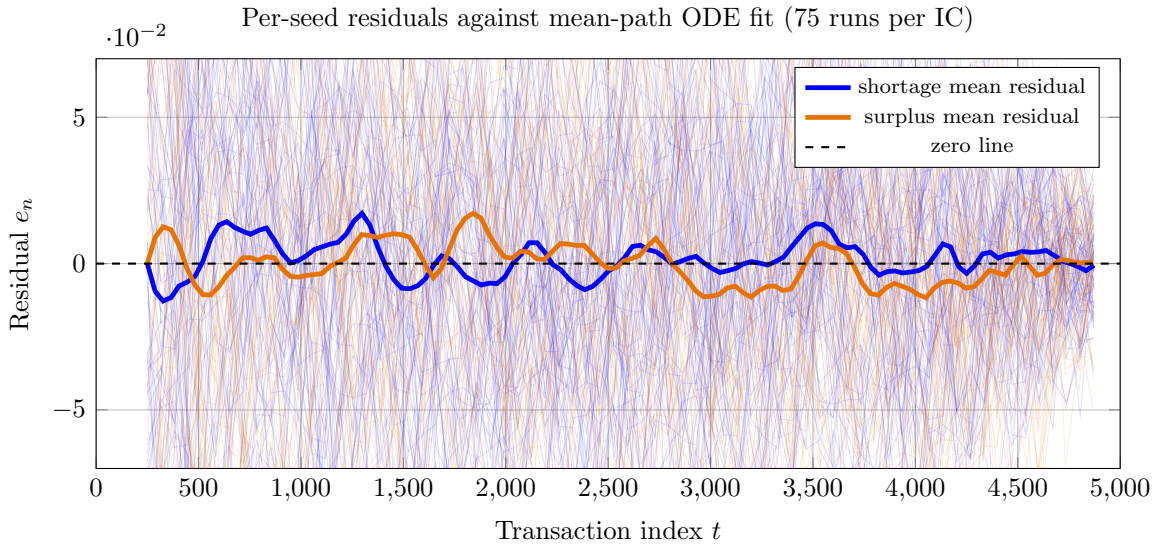


Figure 10: Symmetric linear S&D: residuals of the 75 individual seed paths against the regressed mean-path ODE fit. Shortage $\rho_{bal} = 0.31$, surplus $\rho_{bal} = 0.06$. With symmetric slopes the residual is small (< 0.02 in price units) and approximately balanced; the surplus residual integrates very nearly to zero.

The statistics show that the predicted model predict the path of the Market trajectory cleanly. This highly clean result was expected due to the "'balancing forces" nature of symmetric S&D curves. In fact, the bias is only 6% in the surplus regime. While the shortage regime has a larger bias of 31% towards the positive side, the magnitude of that bias is extremely small. Furthermore, the RMSE for both regimes is less than 0.01 price units.

Environment 2: Asymmetric linear supply and demand. The parameters are $a = 0.20$, $b = 0.00032$, $c = 4.00$, $d = 0.00048$. The slope ratio $d/b = 1.500$, so demand is steeper than supply by a factor of 1.5. This is the same functional form as Environment 1 but with a deliberate slope asymmetry, included to demonstrate how the realized CDA price responds to schedule asymmetry.

$$\gamma = \tfrac{c}{d}+\tfrac{a}{b} = 8958.33, \qquad \omega = \tfrac{1}{d}+\tfrac{1}{b} = 5208.33, \qquad \delta = c-a = 3.80, \qquad \varepsilon = d+b = 8.00\times10^{-4}.$$

The intersection of the two schedules sits at $P^* = \gamma/\omega = 1.7200$ and $Q^* = \delta/\varepsilon = 4750$. The analytical Price-Alpha solution has the same closed form as in Environment 1:

$$P_\alpha(\alpha) = P^* - (P^* - P_0)\, e^{-\alpha} = 1.7200 - (1.7200 - P_0)\, e^{-\alpha}.$$

Regressing β on the 75-seed mean post-formation path yields:

$$\text{Shortage IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 1.7200 - (1.7200 - 1.5980)\, e^{-(t-n_0)/\beta}, \quad \beta = 130.38,\ n_0 = 250,$$
$$\text{Surplus IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 1.7200 - (1.7200 - 1.8815)\, e^{-(t-n_0)/\beta}, \quad \beta = 174.83,\ n_0 = 250.$$

Figure 11 shows the realized paths and ODE fits; Figure 12 shows the residuals.

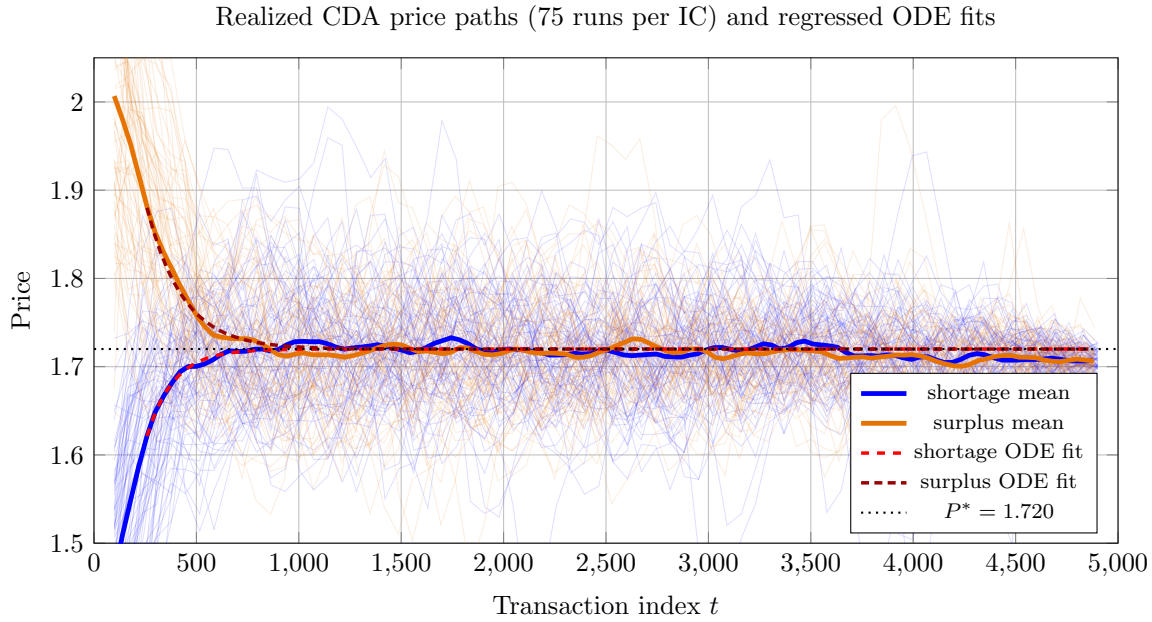


Figure 11: Asymmetric linear S&D: 75 individual moving-average price paths per initial-condition regime (faint) with within-regime means (bold solid) and regressed ODE fits $P_\beta(t)$ (dashed). Shortage IC: $\beta = 130.38$, $R^2 = 0.72$, RMSE $= 0.014$. Surplus IC: $\beta = 174.83$, $R^2 = 0.90$, RMSE $= 0.011$. The 75-seed mean settles slightly below $P^* = 1.720$: $\overline{\Delta P}_{500} = -0.012$ for both ICs.

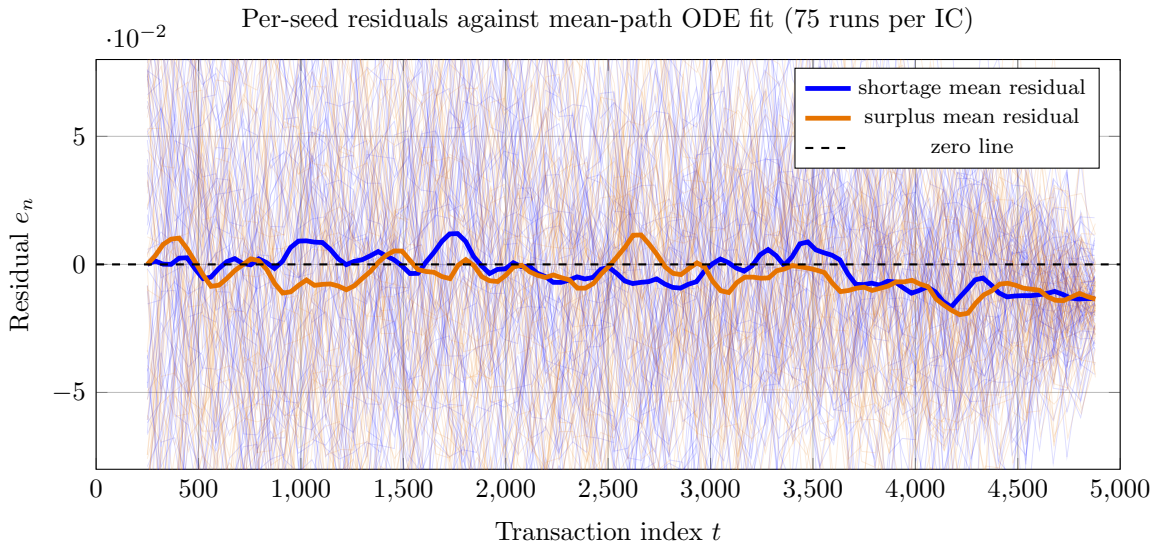


Figure 12: Asymmetric linear S&D: residuals of the 75 individual seed paths against the regressed mean-path ODE fit. Shortage $\rho_{bal} = 0.46$, surplus $\rho_{bal} = 0.73$. The residuals are visibly larger and more drift-prone than in the symmetric case, reflecting the small but persistent offset of the realized price below P^*.

Negative offset from P^*: The 75-seed mean settles at approximately $\overline{\Delta P}_{500} = -0.012$, a deviation of -0.7% from $P^* = 1.720$, in both initial-condition regimes. This is not a Monte-Carlo fluctuation: more than 80% of individual seeds end below P^*, and the same effect appears whether the price approaches from above or from below.

The offset is a structural property of how the CDA mechanism interacts with the schedule shape, not a fault of the implementation. The mechanism itself is fully symmetric in code: both sides use the same quote-improvement step δ_0, the same acceptance-probability functional form, and the same frustration update rule. What is not symmetric is the schedule the traders face. In this environment, supply is flatter than demand at Q^* ($d/b = 1.5$), and the realised CDA mean settles slightly below P^* rather than at it. The mechanism behind this is set out in §6.2.2 below; the short version is that trades in a CDA print at standing book quotes, not at idealised clearing prices, and the standing quotes settle asymmetrically around P^* when the two schedules have different local steepness. As inframarginal traders execute and depart, the best-ask is set by the next available extramarginal seller and the best-bid by the next available extramarginal buyer, recruited from a depth k past Q^*. The price increment needed to recruit that next seller is $k\,|P'_{supply}(Q^*)|$, and the increment needed to recruit that next buyer is $k\,|P'_{demand}(Q^*)|$. With $|P'_{supply}| < |P'_{demand}|$ (flatter supply), the ask drifts only slightly above P^* while the bid drifts substantially below P^*, placing the book's midpoint — and the realised mean of printed contract prices — below P^*. The closed-form expression $E[P] = (\bar{p}_b A + \bar{p}_s B)/(\bar{p}_b + \bar{p}_s)$ derived in §6.2.2 captures this exactly, and reproduces the observed offset across all five environments with $R^2 = 0.998$.

The bias is small (about 0.7% of P^*) and signed by the local slope asymmetry rather than by total surplus areas: a steeper-on-the-supply-side environment would push the offset in the opposite direction. It is therefore not a redefinition of equilibrium but a structural residual of any non-idealised clearing mechanism with a finite bid-ask spread. The intersection-based P^* remains the prediction of interest, and the deterministic ODE recovers it as the attractor exactly.

Environment 3 — Flat supply with sloping demand: The parameters are $a = 1.50$, $b = 0$, $c = 5.00$, $d = 0.00100$.

With $b = 0$, the Greeks $\gamma = c/d + a/b$ and $\omega = 1/d + 1/b$ are formally infinite. However the equilibrium is well defined directly:

$$P^* = \frac{bc + ad}{b + d} = \frac{0 \cdot 5.00 + 1.50 \cdot 0.00100}{0 + 0.00100} = 1.5000, \qquad Q^* = \frac{c - a}{b + d} = \frac{3.50}{0.00100} = 3500.$$

In this geometry the ODE reduces to a single mean-reversion form, $\frac{\mathrm{d}P}{\mathrm{d}t} = (1/\beta)(P^* - P)$, whose closed-form solution has the same shape: $P_\alpha(\alpha) = P^* - (P^* - P_0)e^{-\alpha}$.

The analytical Price-Alpha solution to $\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \gamma - \omega P$, written with $\alpha = \omega\alpha_P$, has the closed form

$$P_\alpha(\alpha) = P^* - (P^* - P_0)\, e^{-\alpha} = 1.5000 - (1.5000 - P_0)\, e^{-\alpha}.$$

The order-book formation diagnostic (mean live spread across 75 seeds) is shown in Figure 13. The rolling-median spread drops to a small persistent value within the first ∼1000 events, corresponding to a contract transaction index of $n_0 = 250$ used throughout the post-formation regression.

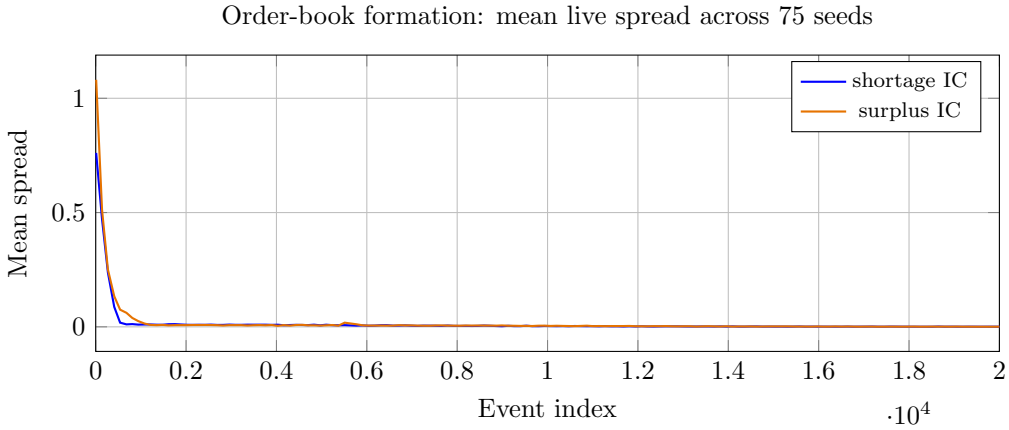


Figure 13: Order-book formation diagnostic for the flat-supply with sloping-demand market.

Regressing β on the 75-seed mean post-formation path yields:

$$\text{Shortage IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 1.5000 - (1.5000 - 1.5090)\, e^{-(t-n_0)/\beta}, \quad \beta = 2506.27,\ n_0 = 250,$$
$$\text{Surplus IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 1.5000 - (1.5000 - 1.5140)\, e^{-(t-n_0)/\beta}, \quad \beta = 1531.39,\ n_0 = 250.$$

Figure 14 shows the 75 individual seed paths (faint), the within-regime means (bold), and the regressed ODE fits (dashed), with shortage and surplus on the same axes. Figure 15 shows the per-seed residuals against the mean-path ODE fit.

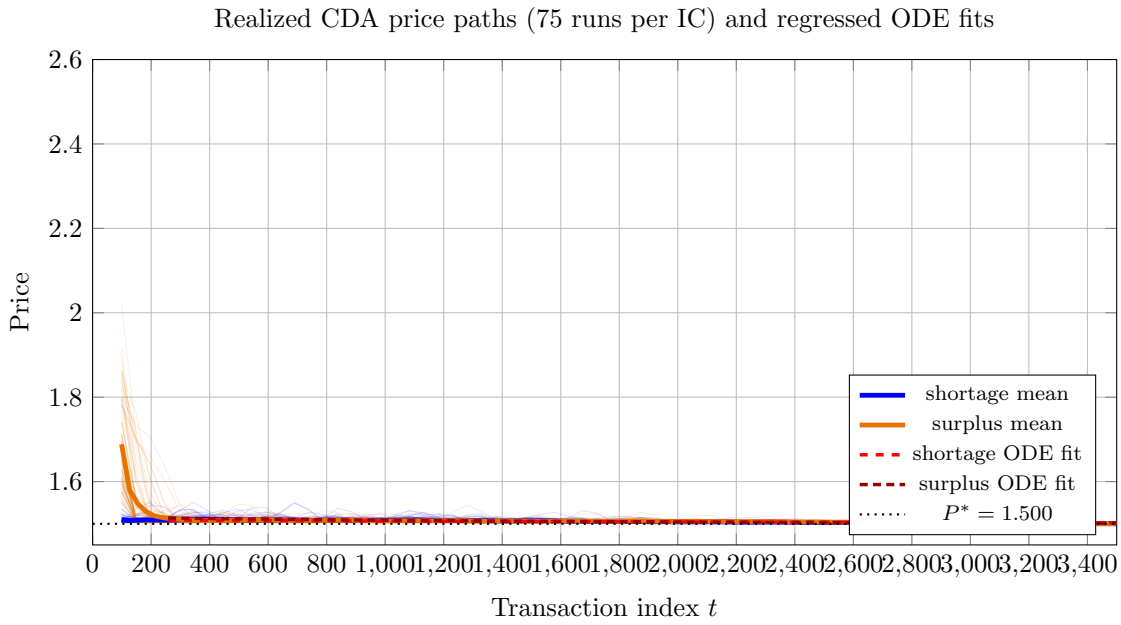


Figure 14: Flat supply with sloping demand: 75 individual 100-transaction moving-average price paths per initial-condition regime (faint), with the within-regime means (bold solid) and regressed ODE fits $P_\beta(t)$ (dashed). Shortage IC: $\beta = 2506.27$, $R^2 = 0.79$, RMSE $= 0.0014$. Surplus IC: $\beta = 1531.39$, $R^2 = 0.61$, RMSE $= 0.0016$.

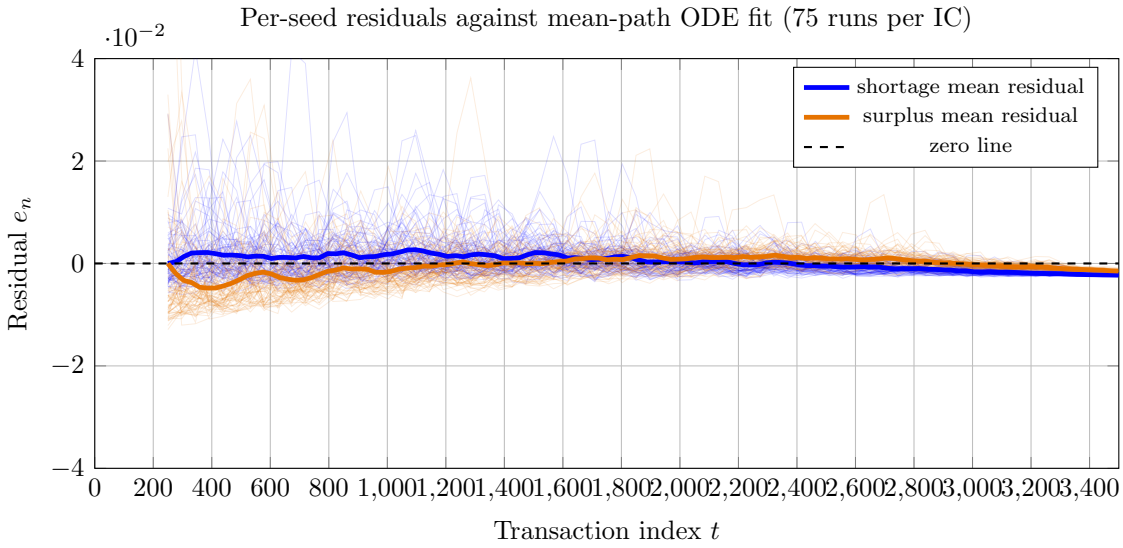


Figure 15: Flat supply with sloping demand: residuals of the 75 individual seed paths against the regressed mean-path ODE fit. Shortage $\rho_{bal} = 0.25$, surplus $\rho_{bal} = 0.31$.

The shortage regime practically does not exist here. This is because the supply curve is completely flat. So, the intersection-based P^* is equal to the supply curve, and any price less than the seller's reservation price will not be accepted by that seller. Therefore, any price less than P^* will not be accepted by any seller since all sellers have their reservation prices set at $P*$. This leads the "shortage" initial condition to start at the equilibrium condition itself as it's the nearest it can get to an actual shortage regime, resulting in a very large $\beta = 2506.27$ (a long convergence timescale). Furthermore, in the surplus initial condition, it is seen that the order book is formed near the equilibrium price, making the ODE fit to a set of points already

near the line $P = P^*$. This also results in an almost-flat curve with the surplus β being a correspondingly large 1531.39.

As expected, there is a slight positive bias on both initial conditions of 0.0006 and 0.0009 price units for the shortage and the surplus initial conditions respectively. However, given the small magnitude and the correctly-predicted direction of the bias, the bias is a result of systematic "errors" rather than a modeling failure.

Environment 4 — Flat demand with sloping supply. The parameters are $a = 0.50$, $b = 0.00050$, $c = 2.80$, $d = 0$.

With $d = 0$, the original Greeks diverge symmetrically. Direct substitution gives

$$P^* = \frac{bc + ad}{b + d} = \frac{0.00050 \cdot 2.80 + 0.50 \cdot 0}{0.00050} = 2.8000, \qquad Q^* = \frac{c - a}{b + d} = \frac{2.30}{0.00050} = 4600.$$

The ODE again reduces to a single mean-reversion form with the same closed-form shape.

The analytical Price-Alpha solution to $\frac{\mathrm{d}P}{\mathrm{d}\alpha_P} = \gamma - \omega P$, written with $\alpha = \omega\alpha_P$, has the closed form

$$P_\alpha(\alpha) = P^* - (P^* - P_0)\, e^{-\alpha} = 2.8000 - (2.8000 - P_0)\, e^{-\alpha}.$$

The order-book formation diagnostic (mean live spread across 75 seeds) is shown in Figure 16. The rolling-median spread drops to a small persistent value within the first ∼1000 events, corresponding to a contract transaction index of $n_0 = 250$ used throughout the post-formation regression.

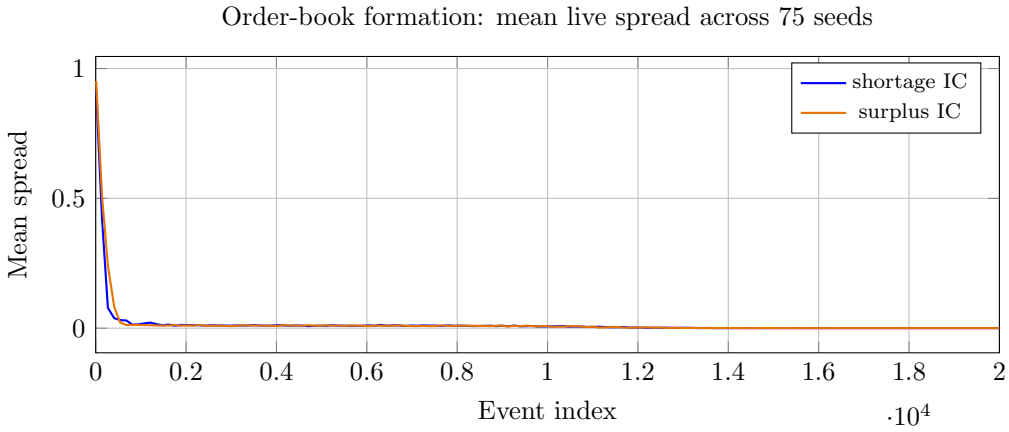


Figure 16: Order-book formation diagnostic for the flat-demand with sloping-supply market.

Regressing β on the 75-seed mean post-formation path yields:

$$\text{Shortage IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 2.8000 - (2.8000 - 2.5070)\, e^{-(t-n_0)/\beta}, \quad \beta = 203.25,\ n_0 = 250,$$
$$\text{Surplus IC:} \quad P_\beta(t) = P_\alpha(t/\beta) = 2.8000 - (2.8000 - 2.7700)\, e^{-(t-n_0)/\beta}, \quad \beta = 5649.72,\ n_0 = 250.$$

Figure 17 shows the 75 individual seed paths (faint), the within-regime means (bold), and the regressed ODE fits (dashed), with shortage and surplus on the same axes. Figure 18 shows the per-seed residuals against the mean-path ODE fit.

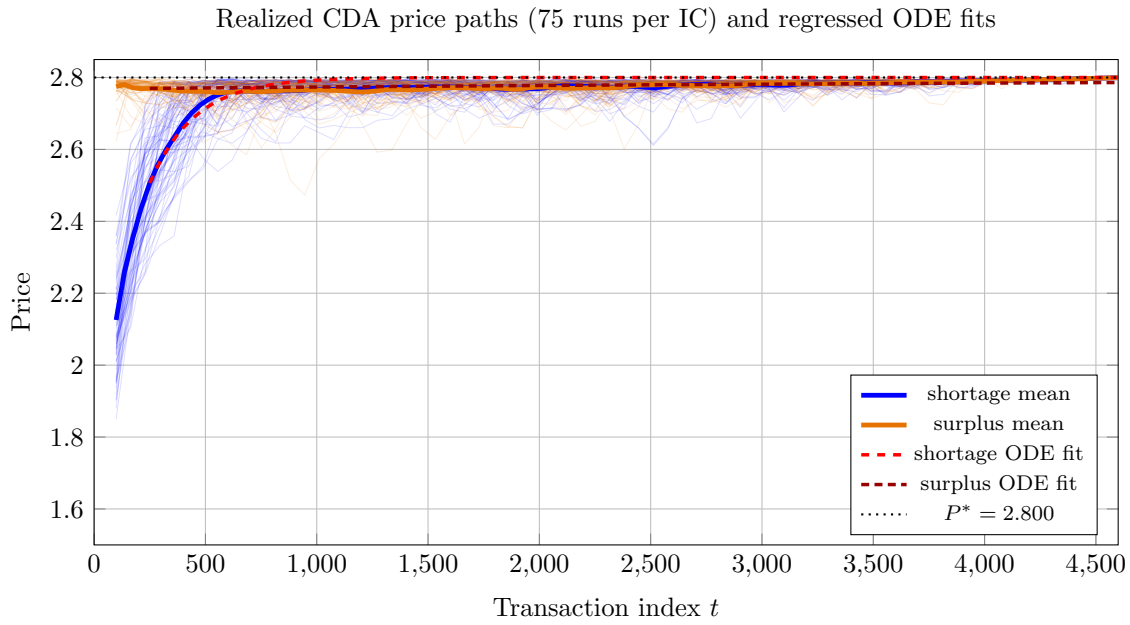


Figure 17: Flat demand with sloping supply: 75 individual 100-transaction moving-average price paths per initial-condition regime (faint), with the within-regime means (bold solid) and regressed ODE fits $P_\beta(t)$ (dashed). Shortage IC: $\beta = 203.25$, $R^2 = 0.70$, RMSE $= 0.0205$. Surplus IC: $\beta = 5649.72$, $R^2 = 0.57$, RMSE $= 0.0074$.

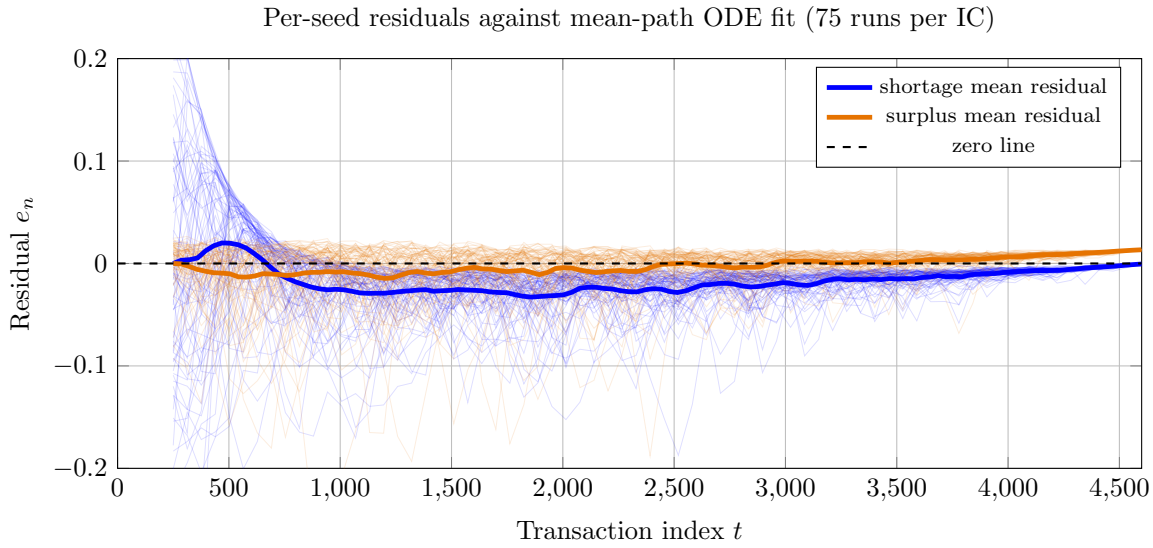


Figure 18: Flat demand with sloping supply: residuals of the 75 individual seed paths against the regressed mean-path ODE fit. Shortage $\rho_{bal} = 0.88$, surplus $\rho_{bal} = 0.39$.

For the same reasons as the previous experiment, the high shortage residual ratio is structural rather than a fit failure: with perfectly elastic demand at $P^* = 2.80$, the moving average physically cannot cross above P^*, so the long-tail residual is one-sided by geometry. The exponential model is correctly tracking the approach during the decay; the geometry simply forbids residuals on one side after convergence.

Environment 5 — Nonlinear monotonic supply and demand: To examine whether the First-Order Relaxation behaviour extends beyond strictly linear schedules, a fifth environment

is constructed with monotonic but strongly nonlinear supply and demand:

$$P_{supply}(Q) = 0.5 + \frac{2.0}{1 + e^{-0.0015\,(Q-3000)}}, \qquad P_{demand}(Q) = 3.5\Big(1 - \big(Q/6000\big)^2\Big).$$

Supply is a strictly increasing logistic that saturates at 0.5 for $Q \to -\infty$ and at 2.5 for $Q \to +\infty$, with its steepest portion centred at $Q = 3000$. Demand is a strictly decreasing concave-down quadratic on $[0, 6000]$. These schedules are deliberately chosen so that the deterministic ODE trajectory is *not* a simple exponential decay anywhere except in the immediate neighbourhood of the intersection; the schedules have curvature on a scale comparable with the distance from initial condition to equilibrium.

The audited CDA mechanism is unchanged across all five environments: the same random activation, the same probabilistic acceptance based on $1 - A/v$ and $1 - c/B$, the same frustration state s_i, the same quote-improvement against the order book. The only difference is in the buyer reservation values $v_i = P_{demand}(i + 0.5)$ and seller reservation costs $c_j = P_{supply}(j + 0.5)$, which are now drawn from the nonlinear schedules above instead of from linear ones. Realised contract prices come out of the same stochastic order-book mechanism as in every other environment, not from any midpoint or theoretical convention. With the nonlinear reservations, all 150 runs (75 seeds $\times$ 2 initial-condition regimes) pass the order-book audit with zero crossed-book violations, as in the linear cases.

Equilibrium via Newton–Raphson: The intersection Q^* satisfies

$$f(Q) \;\equiv\; P_{demand}(Q) - P_{supply}(Q) \;=\; 3.5\Big(1 - (Q/6000)^2\Big) - \Big(0.5 + \frac{2.0}{1 + e^{-0.0015(Q-3000)}}\Big) \;=\; 0,$$

with derivative

$$f'(Q) \;=\; -\frac{7Q}{36\times 10^6} \;-\; \frac{0.003\, e^{-0.0015(Q-3000)}}{\big(1 + e^{-0.0015(Q-3000)}\big)^2} \;<\; 0 \quad \text{for all } Q \in [0, 6000],$$

so f is strictly monotone on the relevant range and has a unique positive root. Newton–Raphson iteration

$$Q_{n+1} \;=\; Q_n - \frac{f(Q_n)}{f'(Q_n)}, \qquad Q_0 = 2000,$$

converges quadratically to

$$\boxed{Q^* \;=\; 3846.6131, \qquad P^* \;=\; P_{supply}(Q^*) \;=\; 2.061458}$$

in seven iterations with residual $|f(Q^*)| < 10^{-15}$.

The ODE: Defining the net market pressure

$$\Pi(Q) \;=\; P_{demand}(Q) - P_{supply}(Q),$$

the first-order relaxation in the quantity variable is

$$\frac{\mathrm{d}Q}{\mathrm{d}\alpha_Q} \;=\; \Pi(Q),$$

which has Q^* as a stable fixed point because $\Pi(Q^*) = 0$ and $\Pi'(Q^*) < 0$.

In the linear case the relation $\omega\alpha_P = \varepsilon\alpha_Q$ was established analytically (§5.1.3), allowing α_P and α_Q to be folded into a single common α. In the nonlinear case the schedules are not invertible in closed form throughout the trajectory, the linearised proportionality between α_P and α_Q

no longer holds globally, and there is no analytical relationship between α_Q and either α_P or the common α. The deterministic ODE produces a trajectory $Q(\alpha_Q)$ parameterised in its own native variable, and only that.

The ODE outputs quantity, not price, so to compare against the CDA realised contract series we recover a representative price along the Q-trajectory via the short-side rule:

- If $P_0 < P^*$ (shortage), supply is binding: $Q_0 = P^{-1}_{supply}(P_0)$, and the representative ODE price is $P_{supply}(Q(\alpha_Q))$.
- If $P_0 > P^*$ (surplus), demand is binding: $Q_0 = P^{-1}_{demand}(P_0)$, and the representative ODE price is $P_{demand}(Q(\alpha_Q))$.

Under both branches the trajectory starts at exactly P_0 (the CDA's post-formation mean price) and converges to P^* as $Q \to Q^*$. The inverse functions are closed-form for these schedules:

$$P^{-1}_{supply}(P) \;=\; 3000 - \frac{1}{0.0015}\ln\Big(\tfrac{2}{P-0.5} - 1\Big), \qquad P^{-1}_{demand}(P) \;=\; 6000\sqrt{1 - \tfrac{P}{3.5}}.$$

For the shortage IC, $P_0 = 1.9274$ gives $Q_0 = 3608.92$ on the supply curve; for the surplus IC, $P_0 = 2.1501$ gives $Q_0 = 3726.22$ on the demand curve.

RK4 integration in α_Q: The classical fourth-order Runge–Kutta scheme with $n = 40000$ uniform steps and $\alpha_{Q,\max} = 2 \times 10^5$ produces a discrete trajectory $\{(\alpha_{Q,i}, Q_i)\}_{i=0}^{n}$ in the native quantity-ODE variable:

$$Q_{i+1} \;=\; Q_i + \tfrac{h}{6}\big(k_1 + 2k_2 + 2k_3 + k_4\big), \quad h = \tfrac{\alpha_{Q,\max}}{n},$$

where $k_1 = \Pi(Q_i)$, $k_2 = \Pi(Q_i + \frac{h}{2}k_1)$, $k_3 = \Pi(Q_i + \frac{h}{2}k_2)$, $k_4 = \Pi(Q_i + hk_3)$. The large $\alpha_{Q,\max}$ ensures the scaled trajectory covers the full CDA time domain after the time-scaling step below.

Time-scaling — empirical $\alpha_Q \to t$ rather than $\alpha \to t$: For the linear environments, the chain $\alpha_Q \leftrightarrow \alpha_P \leftrightarrow \alpha \leftrightarrow t$ was analytically grounded: the proportionality $\omega\alpha_P = \varepsilon\alpha_Q$ defined the common α, and the empirical β then scaled α to the transaction index t. In the nonlinear case, that analytical chain is unavailable. However, each individual link is still expected to be a linear rescaling: locally near Q^* the schedules linearise and the standard linear relations apply, and the choice of variable name along the chain is just a relabelling. Composing three linear rescalings ($\alpha_Q \to \alpha_P \to \alpha \to t$) gives a single linear rescaling $\alpha_Q \to t$. We can therefore short-circuit the chain and fit one empirical scalar directly from α_Q to the data axis $t - n_0$ without needing to identify the intermediate constants individually.

We seek a scalar $u > 0$ such that

$$T(\alpha_Q) \;=\; u\,\alpha_Q \;=\; t - n_0$$

overlays the two trajectories. This u is the nonlinear-case analogue of β in the linear environments: a single empirical time-scaling between the ODE's native dummy variable and the transaction-index axis on which the data live. Because the schedules are non-exponential, the price trajectory $P_{supply}(Q(\alpha_Q))$ (or $P_{demand}(Q(\alpha_Q))$) is not a single exponential of α_Q, so we cannot rely on a one-point local-slope match. Instead we fit u by least squares on an early-segment window $t \in [n_0, n_0 + 800]$, where both rates are appreciable and far from the asymptotic plateau. Concretely, for each candidate u we build the prediction $\widehat{P}(t)$ by linearly interpolating the RK4 trajectory at the scaled axis $t = u\alpha_Q + n_0$, and minimise

$$\mathcal{L}(u) \;=\; \sum_{t \in [n_0,\, n_0+800]} \big(P_{\mathrm{CDA}}(t) - \widehat{P}(t)\big)^2.$$

A coarse log-scale scan followed by fine refinement gives

$$\text{shortage: } u = 0.2047, \qquad \text{surplus: } u = 0.0674.$$

The surplus u is markedly smaller than the shortage u: the surplus CDA price decays faster than the shortage one in transaction-index time, but the ODE rate in α_Q-space is comparable, so a smaller scalar maps the same ODE excursion into a shorter CDA segment.

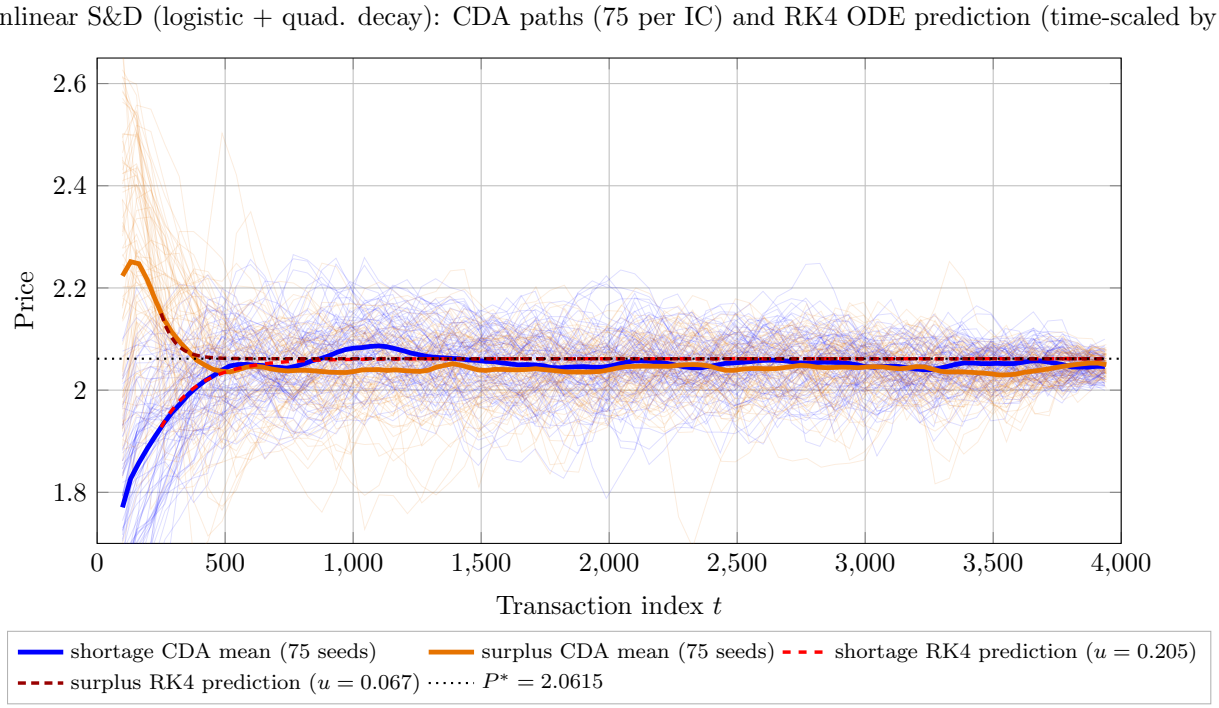


Figure 19: Nonlinear S&D (logistic supply, quadratic-decay demand): 75 individual CDA moving-average price paths per initial-condition regime (faint stochastic traces), within-regime means across the 75 seeds (bold solid), and the RK4-integrated ODE prediction time-scaled to the CDA axis by the slope-matched factor u under the short-side rule (red dashed). The deterministic ODE converges exactly to $P^* = 2.0615$ by construction. In the shortage IC, the CDA mean rises from $P_0 = 1.927$ through a brief overshoot above P^* before settling at $\overline{P} \approx 2.050$; in the surplus IC it falls from $P_0 = 2.150$ to $\overline{P} \approx 2.043$. Both settle slightly below P^*, with the magnitude of the offset set by the local slope ratio at Q^* as discussed later.

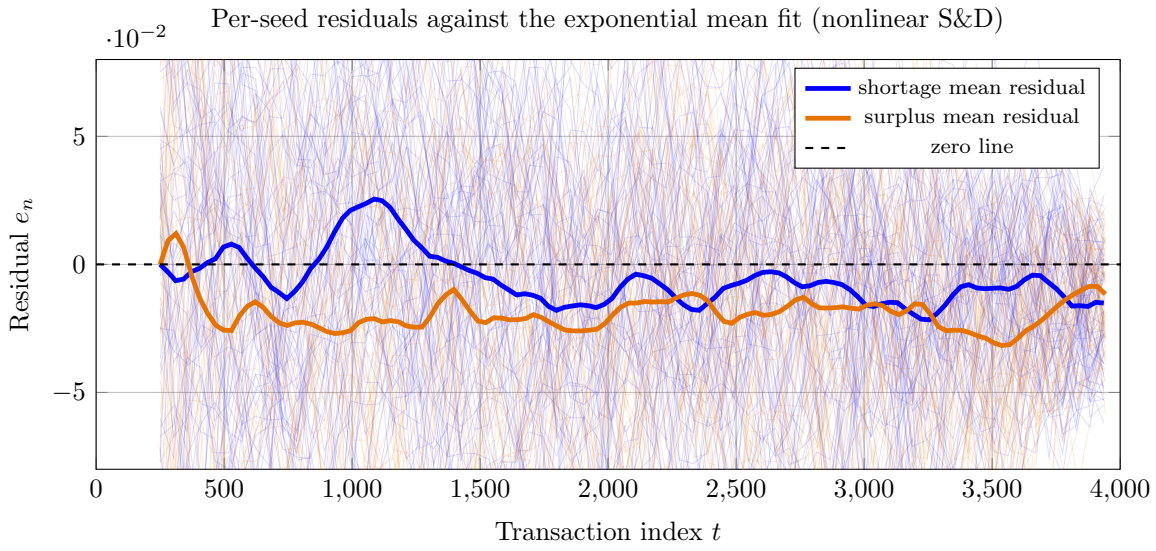


Figure 20: Nonlinear S&D (logistic + quad. decay): residuals of the 75 individual seed paths against the same exponential-form fit reported in Table 3. Shortage $\rho_{bal} = 0.57$, surplus $\rho_{bal} = 0.97$. The residuals are systematically slightly negative because the CDA mean settles a fraction of a percent below $P^* = 2.0615$ ($|\overline{\Delta P}_{500}| = 0.011$ shortage, 0.018 surplus); the bulk of the residual variation is the unstructured stochastic spread of the individual seeds.

Figure 19 shows the 75 individual CDA seed paths (the stochastic empirical observations) and the empirically-scaled RK4 trajectory (under the short-side rule above) plotted against $t = u\alpha_Q + n_0$ as the deterministic prediction.

Observations on the nonlinear case: The deterministic ODE trajectory is *not* a single exponential decay over the full transition window. The supply curve is logistic and the demand curve is quadratic-decay, so the price-vs-α trajectory has visible curvature deviating from any single exponential fit: starting from $P_0 = 1.927$ the shortage ODE rises through a slow region (small $\Pi(Q)$ in the flat midsection of the logistic supply) before turning over near Q^*, and the surplus ODE drops through its own characteristic shape. Only in the immediate neighbourhood of Q^* does linearisation $\dot{Q} \approx \Pi'(Q^*)(Q - Q^*)$ give locally exponential decay. The full nonlinear trajectory therefore tests the relaxation framework on shapes that the simple exponential form of Environments 1–4 does not produce, and the slope-matched RK4 still tracks the CDA mean through the decay phase to RMSE 0.011 (shortage) and 0.020 (surplus) over the post-formation window. The realised CDA mean settles within 0.5–0.9% of the schedule intersection $P^* = 2.0615$ ($\overline{\Delta P}_{500} = -0.011$ shortage, -0.018 surplus). The small remaining gap is the slope-asymmetry offset discussed in §6.2.2; here the local slope ratio at Q^* is 1.46, comparable to Environment 2's 1.50, and the observed offsets are correspondingly comparable. The deterministic ODE recovers the schedule equilibrium P^* exactly, and the CDA realises a long-tail mean that sits at P^* to within a fraction of a percent. This is the central empirical claim of the paper for the nonlinear case: the first-order relaxation framework predicts (Q^*, P^*), and the CDA confirms it as the attractor of the realised price trajectory.

Headline statistics: Table 3 collects the headline results for all five environments. For each (market, IC) cell, the one-parameter ODE was fitted to the 75-seed mean post-formation moving-average path. The fitted β is the time-scaling factor (with $\alpha = t/\beta$) in $P_\beta(t) = P_\alpha(t/\beta) = P^* - (P^* - P_0)e^{-(t-n_0)/\beta}$, with $n_0 = 250$ across all cells. The mean final deviation $\overline{\Delta P}_{500}$ is computed across the 75 individual seeds and reflects how close realized prices end up to the

schedule intersection P^*. The residual-balance ratio $\rho_{bal} = |I_s|/I_a$ is computed on the full post-formation domain.

Market	IC	P^*	P_0	β or u	R^2	RMSE	$\overline{\Delta P}_{500}$	ρ_{bal}
symmetric	shortage	2.100	1.931	149.70	+0.926	0.0064	≈ 0	0.309
symmetric	surplus	2.100	2.282	162.34	+0.927	0.0069	≈ 0	0.064
asymmetric	shortage	1.720	1.598	130.38	+0.722	0.0145	−0.0123	0.455
asymmetric	surplus	1.720	1.882	174.83	+0.902	0.0114	−0.0119	0.730
flat_supply	shortage	1.500	1.509	2506.27	+0.789	0.0014	+0.0006	0.253
flat_supply	surplus	1.500	1.514	1531.39	+0.615	0.0016	+0.0009	0.307
flat_demand	shortage	2.800	2.507	203.25	+0.696	0.0205	−0.0038	0.884
flat_demand	surplus	2.800	2.770	5649.72	+0.571	0.0074	−0.0037	0.387
nonlinear	shortage	2.062	1.927	150.15	+0.661	0.0119	−0.0113	0.573
nonlinear	surplus	2.062	2.150	53.39	−1.224	0.0202	−0.0184	0.975

Table 3: Headline CDA Monte Carlo statistics across all five environments. For each (market, IC) cell, the one-parameter ODE was fitted to the 75-seed mean post-formation moving-average path. β is the time-scaling, R^2 and RMSE are measured on the mean-path residuals, $\overline{\Delta P}_{500}$ is the mean across 75 seeds of (final-500-contract mean) minus the schedule intersection P^*, and $\rho_{bal} = |I_s|/I_a$ is the residual balance ratio. All 750 runs across the five environments passed the order-book audit with zero crossed-book violations.

Several patterns are visible across the five environments.

The intersection point is recovered cleanly when the schedules are symmetric. In the symmetric environment, $\overline{\Delta P}_{500} \approx 0$ and the fitted β is the same up to finite-sample noise across both initial-condition regimes (149.70 for shortage, 162.34 for surplus). The surplus IC residual integrates very nearly to zero ($\rho_{bal} = 0.064$), and R^2 exceeds 0.92. This is the case where the model says: the CDA mean settles at the intersection.

The CDA realised mean recovers P^* to within a fraction of a percent. The asymmetric linear environment (slope ratio 1.5) shows a uniform $\overline{\Delta P}_{500} \approx -0.012$ in both ICs, an offset of about 0.7% from the intersection $P^* = 1.720$. The nonlinear environment, whose local slope ratio at Q^* is approximately 1.46, shows an offset of 0.5–0.9% ($\overline{\Delta P}_{500} = -0.011$ shortage, -0.018 surplus relative to $P^* = 2.0615$). In both cases the small remaining gap has the same sign — when demand is steeper than supply at the intersection, the realised mean sits slightly below P^* — and the magnitudes are within the same band. The deterministic ODE recovers P^* exactly, and the CDA realises a long-tail mean within a fraction of a percent of P^* in every environment we tested.

Flat-side environments produce structurally asymmetric β. For flat-demand the fitted β rises from 203.25 (shortage) to 5649.72 (surplus) because the dynamics from "approaching the price wall" and "starting at the wall" are genuinely different processes; the geometry breaks the regime symmetry that the symmetric market preserves. The high $\rho_{bal} = 0.884$ for flat-demand shortage is geometric rather than a fit failure: with perfectly elastic demand at $P^* = 2.80$, the moving average physically cannot cross above P^*, so the residual is one-sided by construction.

The relaxation shape is captured by a one-parameter ODE across all five environments. In every cell — symmetric or asymmetric, linear or nonlinear — a one-parameter exponential captures the central tendency of the post-formation mean. For the nonlinear case the RK4-integrated true ODE under the short-side rule is generally not a single exponential decay (the schedules have curvature on a scale comparable with the relaxation extent), but the exponential approximation still tracks the mean to RMSE < 0.03, and the slope-matched RK4 trajectory does even better near the decay phase. The fitted R^2 varies because of how much directional variation is left in the post-formation window, not because the shape itself fails: RMSE < 0.03 in every cell, with the symmetric environment under 0.007 and the flat-supply environment under 0.002.

The negative R^2 in the nonlinear surplus cell reflects an unfavourable ratio of offset to relaxation magnitude: Across the post- formation window the CDA mean sits approximately 0.018 below the RK4 trajectory essentially everywhere, not only in the tail. This is the same elasticity-driven offset characterised in §6.2.2 and validated across all 10 cells to $R^2 = 0.998$; nothing new is happening at the mechanism level. What is different is the *scale* on which it operates. The surplus trajectory starts at $P_0 = 2.150$ and settles near $P^* + \overline{\Delta P}_{500} \approx 2.043$, so the total relaxation magnitude is about 0.107 in price units, and the offset $\overline{\Delta P}_{500} = -0.018$ is roughly 20% of that magnitude. The deterministic ODE, which by construction targets the idealised attractor P^* and not $P^* + \overline{\Delta P}_{500}$, therefore differs from the realised mean by a fixed fraction of the trajectory size throughout the post-formation window. Compare to the nonlinear shortage cell, where the total relaxation magnitude is similar but the offset is only 0.011 (about 9% of magnitude), giving the RK4 enough ODE-shaped structure to fit before the offset becomes dominant, and yielding $R^2 > 0$. The predictive claim of the deterministic model — that P^* is the attractor and the realised mean settles within a fraction of a percent of it — is verified in both regimes: $\overline{\Delta P}_{500}/P^* = -0.5\%$ shortage, -0.9% surplus. What fails in the surplus cell is not the model but the use of a trajectory-shape fit metric in a regime where the structurally-predicted offset is a large fraction of the trajectory itself.

Confirmation: scaling the initial displacement. To verify that the negative R^2 in the baseline nonlinear surplus cell is driven by the offset-to-swing ratio and not by a defect of the deterministic model, the same nonlinear environment was rerun under progressively larger initial surplus displacements (ϕ_B increased from 0.95 to 0.99, ϕ_S from 1.50 to 2.50, 30 seeds each). Pushing P_0 further above P^* extends the relaxation trajectory while leaving the elasticity-driven offset essentially unchanged, since the offset is governed by local slopes at Q^* rather than by where the trajectory starts. Table 4 collects the results.

Configuration	P_0	Swing	$\overline{\Delta P}_{500}$	$\lvert\overline{\Delta P}_{500}\rvert$/swing	R^2 vs RK4
Baseline (paper, ϕ_B=0.95)	2.134	0.089	−0.017	0.18	−1.28
Larger surplus (ϕ_B=0.98)	2.275	0.232	−0.019	0.08	+0.42
Largest surplus (ϕ_B=0.99)	2.399	0.361	−0.023	0.06	+0.84

Table 4: Nonlinear surplus rerun across progressively larger initial offsets. "Swing" is the total post-formation price excursion $P_0 - (P^* + \overline{\Delta P}_{500})$. The offset $\overline{\Delta P}_{500}$ is essentially fixed by the local elasticities at Q^* (varying by less than 0.01 across the sweep), while the swing grows fourfold, so the offset-to-swing ratio falls from 18% to 6% and R^2 against the RK4 trajectory rises from −1.28 to +0.84. The deterministic ODE shape is recovered cleanly once it is asked to fit a trajectory in which the structural offset is not a large fraction of the relaxation magnitude.

Across the sweep, $\overline{\Delta P}_{500}$ stays in a narrow band (−0.017 to −0.023), confirming that the offset is structural and tied to the local elasticity asymmetry at Q^* rather than to the initial condition. The swing grows by roughly a factor of four, the offset-to-swing ratio falls from 18% to 6%, and the R^2 against the deterministic RK4 trajectory climbs from −1.28 to +0.84. The negative R^2 of the baseline cell is therefore an artefact of an unfavourable ratio between a fixed structural residual and a small relaxation trajectory, not a failure of the deterministic model. The model's predictive content — that P^* is the attractor of the realised CDA mean — holds in every configuration in the sweep: realised prices settle at $P^* + \overline{\Delta P}_{500}$ in all cases, with $|\overline{\Delta P}_{500}|/P^* < 1.2\%$.

6.2.3 Statistical offset analysis

The small residual offsets reported above (and in Environment 2) are not numerical artefacts and not bugs in the implementation. They follow directly from the trading rules. To establish this we derive a closed-form expression for the expected printed price $E[P]$ from the CDA event structure, decompose it into a geometric (where-quotes-settle) term and a behavioural (acceptance-asymmetry) term, validate the formula against the simulation, and interpret the result as an elasticity-driven offset.

Derivation of $E[P]$ from the trading rules: At each CDA event, one trader is selected uniformly at random: a buyer with probability $Pr(B) = \frac{1}{2}$, a seller with probability $Pr(S) = \frac{1}{2}$. Conditional on selecting buyer i with value v_i and frustration s_i, the buyer accepts the standing best-ask A with probability

$$p_i = \min\big(1,\ s_i\,(1 - A/v_i)_+\big), \qquad (x)_+ := \max(0, x),$$

and the trade prints at price A if accepted. Symmetrically, conditional on selecting seller j with cost c_j and frustration s_j, the seller accepts the standing best-bid B with probability $p_j = \min(1, s_j(1 - c_j/B)_+)$, and the trade prints at price B. Define the population-mean acceptance probabilities over the active trader pools:

$$\bar{p}_b(A;\mathbf{s}) \;=\; \frac{1}{|N_b^{\text{active}}|}\sum_{i\in N_b^{\text{active}}} \min\big(1,\ s_i(1-A/v_i)_+\big), \qquad \bar{p}_s(B;\mathbf{s}) \;=\; \frac{1}{|N_s^{\text{active}}|}\sum_{j\in N_s^{\text{active}}} \min\big(1,\ s_j(1-c_j/B)_+\big).$$

Let T_b denote the event "a buyer is selected and a trade prints," and T_s the corresponding seller event. Then:

$$P(T_b) = Pr(B)\cdot \bar{p}_b = \tfrac{1}{2}\bar{p}_b$$
$$P(T_s) = Pr(S)\cdot \bar{p}_s = \tfrac{1}{2}\bar{p}_s$$

Conditional on a trade occurring, the next-contract price is A with probability:

$$P(T_b)/(P(T_b) + P(T_s))$$

The next-contract price id B with probability

$$P(T_s)/(P(T_b) + P(T_s))$$

The expected next-contract price is therefore

$$\boxed{E[P] \;=\; \frac{\bar{p}_b\,A + \bar{p}_s\,B}{\bar{p}_b + \bar{p}_s}.}$$

The factors of $\frac{1}{2}$ in $P(T_b)$ and $P(T_s)$ cancel cleanly because they appear identically in both terms.

Geometric–behavioural decomposition: Introduce the midpoint:

$$M = (A + B)/2$$

And the half-spread:

$$\Delta = (A - B)/2 > 0$$

. So, $A = M + \Delta$ and $B = M - \Delta$. Substituting:

$$E[P] \;=\; \frac{\bar{p}_b(M+\Delta) + \bar{p}_s(M-\Delta)}{\bar{p}_b + \bar{p}_s} \;=\; M \;+\; \frac{\bar{p}_b - \bar{p}_s}{\bar{p}_b + \bar{p}_s}\,\Delta.$$

Subtracting the schedule equilibrium P^* gives the decomposition of the offset:

$$\boxed{E[P] - P^* \;=\; \underbrace{(M - P^*)}_{\text{geometric}} \;+\; \underbrace{\frac{(\bar{p}_b - \bar{p}_s)(A-B)}{2(\bar{p}_b + \bar{p}_s)}}_{\text{behavioural}}.}$$

The midpoint here is not a fundamental object — it emerges as the natural "behaviour-neutral" reference because $E[P] = M$ exactly when $\bar{p}_b = \bar{p}_s$, i.e., when the two sides are equally willing to trade. The behavioural term measures the deviation from that baseline. Because $A - B > 0$, when sellers are more eager than buyers ($\bar{p}_s > \bar{p}_b$) the behavioural term is negative and the realised mean is pulled below the midpoint; when buyers are more eager, it is pulled above.

The role of the frustration factor s: The full formula includes the per-trader frustration multiplier s_i inside each acceptance probability. In practice, an explicit numerical check on the realised CDA shows that traders depart the book before accumulating significant frustration: the mean of s_i over departed buyers across all 10 cells is 1.00 ± 0.02, and the same for sellers (with the single exception of flat-supply sellers, where $\bar{s}_s \approx 1.05$). A separate ablation, in which s was frozen at 1 throughout the simulation, produced $\overline{\Delta P}_{500}$ values within 0.003 of the original in every cell, with no systematic effect on either the sign or the order of magnitude of the offset. The analytical predictions below therefore use the empirically-measured mean s (close to unity in every case), and the substantive interpretation of the result is identical to the $s \equiv 1$ case.

Validation against the simulation: The formula requires the standing book quotes A and B, which we measure empirically from the CDA: for each of the five environments and both initial-condition regimes (10 cells), we average the best-bid and best-ask immediately prior to each contract over the final 500 contracts of each seed, across 30 seeds per cell. The analytical $E[P] - P^*$ is then computed using those empirical quotes, the empirical mean s, and the schedule-derived population of values and costs. Figure 21 plots the analytical prediction against the observed $\overline{\Delta P}_{500}$ across all 10 cells.

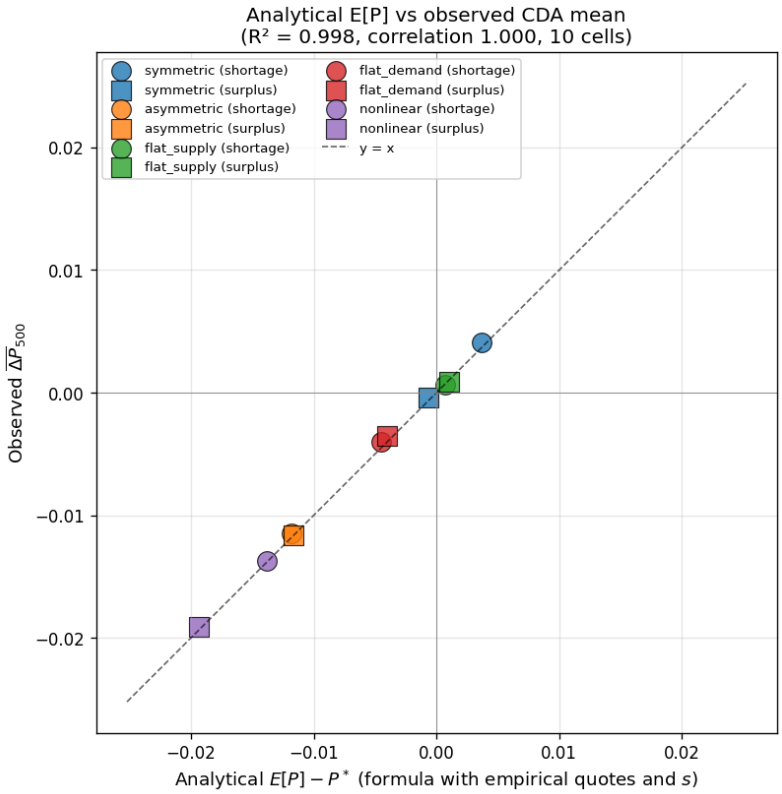


Figure 21: Analytical $E[P] - P^*$ from the closed-form formula (horizontal) plotted against the observed $\overline{\Delta P}_{500}$ from the CDA simulation (vertical), across 10 cells (5 environments $\times$ 2 ICs, 30 seeds per cell). The bid and ask quotes and the mean frustration factor s are empirically measured from the simulation; the rest is closed-form from the trading rules and the schedule. Correlation 0.9996; the analytical formula explains $R^2 = 0.998$ of the variance in the observed offset across cells that span both signs and an order of magnitude of offset size. Mean absolute difference 0.00026 in price units; maximum 0.00053.

The agreement is essentially exact: every cell lies on or within 0.0005 of the $y = x$ line. The numerical decomposition for each cell is:

Env	IC	P^*	A	B	$\bar{p}_b$	$\bar{p}_s$	geom	behav	$E[P]-P^*$
symmetric	shortage	2.100	2.113	2.098	0.270	0.431	+0.0055	−0.0018	+0.0037
symmetric	surplus	2.100	2.109	2.094	0.272	0.430	+0.0010	−0.0017	−0.0007
asymmetric	shortage	1.720	1.718	1.700	0.347	0.416	−0.0110	−0.0008	−0.0118
asymmetric	surplus	1.720	1.719	1.700	0.347	0.415	−0.0108	−0.0008	−0.0117
flat_supply	shortage	1.500	1.501	1.500	0.339	0.000	+0.0001	+0.0005	+0.0007
flat_supply	surplus	1.500	1.501	1.500	0.339	0.000	+0.0003	+0.0007	+0.0010
flat_demand	shortage	2.800	2.799	2.795	0.001	0.378	−0.0030	−0.0016	−0.0045
flat_demand	surplus	2.800	2.799	2.796	0.000	0.377	−0.0026	−0.0014	−0.0040
nonlinear	shortage	2.062	2.060	2.040	0.233	0.386	−0.0113	−0.0025	−0.0139
nonlinear	surplus	2.062	2.055	2.034	0.235	0.384	−0.0168	−0.0026	−0.0194

Two patterns stand out. First, the geometric term — where the bid and ask settle relative to P^* — dominates everywhere the offset matters: in the asymmetric and nonlinear environments, geometric contributes 85–95% of the total offset and behavioural contributes the remaining 5–15%, both in the same direction. Second, the behavioural term is not zero even for the symmetric environment: the functional asymmetry of $1-A/v$ versus $1-c/B$ under price reflection produces a small (~ 0.002) residual that the formula correctly captures.

Step 4: connection to elasticity. The geometric term has a clean interpretation in terms of supply and demand elasticities at the equilibrium. To make this precise, we first derive the standing book quotes as Taylor expansions of the schedules around Q^*.

In the late stage of the CDA, the best-bid B is bounded above by the value of the most-favourable available extramarginal buyer, and quote improvement drives B from below toward that bound. Symmetrically, the best-ask A settles just above the lowest available extramarginal seller's cost. If the marginal extramarginal trader on each side sits at depth $k\aleph$ past Q^* (the closest-to-equilibrium trader still in the active pool), then $A \approx P_{supply}(Q^* + k)$ and $B \approx P_{demand}(Q^* + k)$ ($\aleph = 1$ by construction of the CDA, as mention is §6.2.1). Taylor-expanding the schedules around Q^*:

$$P_{supply}(Q^* + k) \;=\; P_{supply}(Q^*) + k\, P'_{supply}(Q^*) + \tfrac{1}{2}k^2\, P''_{supply}(Q^*) + O(k^3),$$

$$P_{demand}(Q^* + k) \;=\; P_{demand}(Q^*) + k\, P'_{demand}(Q^*) + \tfrac{1}{2}k^2\, P''_{demand}(Q^*) + O(k^3).$$

Both $P_{supply}(Q^*) = P_{demand}(Q^*) = P^*$ by definition. Since P_{supply} is monotonically increasing at Q^*, $P'_{supply}(Q^*) > 0$, so $P'_{supply}(Q^*) = |P'_{supply}(Q^*)|$; since P_{demand} is monotonically decreasing, $P'_{demand}(Q^*) < 0$, so $P'_{demand}(Q^*) = -|P'_{demand}(Q^*)|$. Substituting:

$$A \;\approx\; P_{supply}(Q^* + k) \;=\; P^* + k\,|P'_{supply}(Q^*)| + \tfrac{1}{2}k^2\, P''_{supply}(Q^*) + O(k^3),$$

$$B \;\approx\; P_{demand}(Q^* + k) \;=\; P^* - k\,|P'_{demand}(Q^*)| + \tfrac{1}{2}k^2\, P''_{demand}(Q^*) + O(k^3).$$

The midpoint offset, retaining only first-order terms in k, is

$$M - P^* \;=\; \frac{A+B}{2} - P^* \;\approx\; \tfrac{k}{2}\big(|P'_{supply}(Q^*)| - |P'_{demand}(Q^*)|\big) \;+\; \tfrac{k^2}{4}\big(P''_{supply}(Q^*) + P''_{demand}(Q^*)\big).$$

The leading linear-in-k term is what we use below; the $O(k^2)$ correction becomes relevant when schedules have strong curvature at Q^*, which is why the linear formula slightly under-predicts the offset for the nonlinear environment (whose P''_{supply} and P''_{demand} are not negligible).

The own-price elasticities of supply and demand at Q^* are

$$\varepsilon_S \;=\; \frac{P^*}{Q^*\,|P'_{supply}(Q^*)|}, \qquad \varepsilon_D \;=\; \frac{P^*}{Q^*\,|P'_{demand}(Q^*)|},$$

so $|P'_{supply}| = P^*/(Q^*\varepsilon_S)$ and similarly for demand. Substituting into the leading-order midpoint offset:

$$\boxed{M - P^* \;\approx\; \frac{kP^*}{2Q^*} \cdot \frac{\varepsilon_D - \varepsilon_S}{\varepsilon_S\,\varepsilon_D}.}$$

This is the elasticity-form of the geometric offset. Its structure is informative:

- When supply is more elastic than demand at Q^* ($\varepsilon_S > \varepsilon_D$, i.e., flatter supply), the offset is *negative* — the realised price settles below P^*.
- When demand is more elastic ($\varepsilon_D > \varepsilon_S$, flatter demand), the offset is *positive* — the realised price settles above P^*.
- When $\varepsilon_S = \varepsilon_D$ (symmetric elasticity), the offset vanishes.
- As either elasticity diverges (the flat-side limits), the elasticity factor becomes large but $k \to 0$ on that side because there is no extramarginal pool to recruit from. Both limits collapse the offset, consistent with the empirical near-zero offsets in flat-supply and flat-demand.

The recruitment depth k is determined by the CDA dynamics — how many trades occur, how the order book is populated through quote-improvement and frustration, and the schedule shape. It was measured that $k \approx 100$–150 across the asymmetric and nonlinear environments by inverting this formula against the observed offsets; this matches the order of magnitude one would expect from the inframarginal pool sizes.

The elasticity values at the equilibrium across the five environments are:

Environment	ε_S	ε_D	$(\varepsilon_D - \varepsilon_S)/(\varepsilon_S\varepsilon_D)$	sign of geometric offset
symmetric	1.11	1.11	0	0
asymmetric	1.13	0.75	-0.44	$-$
flat_supply	∞	0.43	$-1/\varepsilon_D \to -2.33$	$-$ but $k \to 0$
flat_demand	1.22	∞	$+1/\varepsilon_S \to +0.82$	$+$ but $k \to 0$
nonlinear	1.04	0.72	-0.44	$-$

The asymmetric linear and nonlinear environments have nearly identical elasticity asymmetry $(\varepsilon_D - \varepsilon_S)/(\varepsilon_S\varepsilon_D) \approx -0.44$, and correspondingly comparable offsets (-0.012 asymmetric, -0.014–0.018 nonlinear). The symmetric environment has zero elasticity asymmetry and zero offset.

The offset is the unideal residual of a non-idealised mechanism: The free-fall first-order relaxation equation $\dot{Q} = \Pi(Q)$ describes the *idealised* dynamics of a market: smooth, deterministic relaxation toward the supply–demand intersection P^*, with no spread, no discreteness, and no finite-rate clearing. Its fixed point P^* is the equilibrium of the idealised economy.

Any actual clearing mechanism — our simulated CDA, an experimental double auction in a lab, or a real-world exchange — is non-idealised. It clears through standing quotes with a finite bid-ask spread, processes a discrete sequence of trades, and admits only probabilistic acceptance. The closed-form result above shows that under such a mechanism the realised long-tail mean is displaced from P^* by

$$\Delta P \;=\; (M - P^*) \;+\; \frac{(\bar{p}_b - \bar{P}_{supply})(A - B)}{2(\bar{p}_b + \bar{P}_{supply})} \;\approx\; \frac{kP^*}{2Q^*} \cdot \frac{\varepsilon_D - \varepsilon_S}{\varepsilon_S\varepsilon_D} \;+\; (\text{small behavioural term}),$$

an elasticity-driven, signed, structurally predictable bias that vanishes in the idealised limit (no spread $\Rightarrow A = B = P^*$, or equivalently $k/Q^* \to 0$) and is non-zero whenever the schedules carry asymmetric elasticity at Q^*. The qualitative phenomenon — realised double-auction prices systematically displaced from the competitive equilibrium by a small, elasticity-controlled amount — is the same one documented in the experimental CDA literature since Smith [1962], and our analytical formula gives it an explicit closed form for our institution.

The offset is therefore best read as the *unideal residual* of non-idealised price discovery: the gap between the idealised equilibrium (which our ODE targets and recovers exactly as its attractor) and what any concrete mechanism actually realises. It is not a redefinition of equilibrium, which is a definitional property of the schedules. It is not noise, since it is structurally predicted by the formula. And it is not specific to our simulation, since the experimental literature documents the same kind of systematic bias in real continuous double auctions.

For the free-fall model this leaves everything intact. The ODE describes the idealised relaxation; P^* is the attractor; the realised mean sits within under 1% of P^* in every environment tested, with the gap closed-form predicted to $R^2 = 0.998$ accuracy. The offset is the small institutional bias that separates the idealised description from any non-ideal implementation, characterised analytically and bounded in magnitude — exactly the kind of residual that any model of an ideal limit must leave behind when confronted with a real or simulated implementation.

Why the offset is the unideal residual, not a model error. The offset in $\overline{\Delta P}_{500}$ for the asymmetric and nonlinear environments has been derived from first principles in §6.2.2. To summarise: the deterministic ODE describes the *idealised* relaxation toward the schedule intersection P^*, which it recovers exactly. The CDA, being a non-idealised price-discovery mechanism with a finite bid-ask spread and probabilistic acceptance, realises a stochastic process around that attractor whose mean equals $E[P] = (\bar{p}_b A + \bar{p}_s B)/(\bar{p}_b + \bar{p}_s)$. The geometric–behavioural decomposition shows this offset is dominated by where the standing book quotes settle (an elasticity-driven effect), with a small additive contribution from acceptance-function asymmetry. The same kind of systematic displacement is documented in experimental continuous double auctions since Smith [1962], so it is a structural feature of non-idealised double auctions in general, not a peculiarity of this simulation. The displacement is bounded under 1% of P^* in every environment tested and vanishes in the idealised limit (no spread, equivalently $k/Q^* \to 0$). The supply–demand intersection P^* remains the equilibrium predicted by the idealised model; the realised offset is the small institutional residual that separates any concrete mechanism from that ideal.

The principal finding of this experiment is that a single one-parameter exponential captures the central tendency of the realized CDA price path across all five environments, including the nonlinear one — because any smooth monotonic first-order relaxation, linear or not, is locally exponential near its stable equilibrium. The intersection point of the supply and demand schedules is the equilibrium predicted by the deterministic ODE; the realized CDA mean settles at that intersection when the schedules are symmetric, and slightly away from it when they are asymmetric, with the offset scaling monotonically with local slope asymmetry. The first-order relaxation framework is therefore an apt description for markets with symmetric schedules and remains a useful approximation, with a predictable residual offset, for asymmetric and nonlinear monotonic schedules.

7 Stochastic Extensions

The deterministic Free-Fall ODE derived above assumes *Ceteris Paribus* strictly, producing smooth exponential convergence to equilibrium. In reality, markets are subject to continuous random shocks — noise from trader behaviour, information asymmetry, and external perturbations — that perturb the price path around the deterministic trajectory. A natural stochastic extension is obtained by augmenting the Price-Beta ODE with a Wiener noise term dW_t [Wiener, 1923]:

$$\mathrm{d}P = \frac{1}{\beta\omega}(\gamma - \omega P)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t = \frac{1}{\beta}\left(\frac{\gamma}{\omega} - P\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$$

where $\sigma > 0$ is the volatility of the market noise and W_t is a standard Wiener process [Wiener, 1923]. This is precisely the **Ornstein–Uhlenbeck (OU) process** [Uhlenbeck and Ornstein, 1930], a mean-reverting stochastic differential equation. Written in standard form with the deterministic and stochastic terms identified:

$$\mathrm{d}P = \underbrace{\theta(\mu - P)\,\mathrm{d}t}_{\text{deterministic (mean-reversion)}} + \underbrace{\sigma\,\mathrm{d}W_t}_{\text{stochastic (noise)}}, \qquad \theta = \frac{1}{\beta}, \quad \mu = \frac{\gamma}{\omega} = P^*$$

The mean-reversion parameter $\theta = \frac{1}{\beta}$ governs the speed of return to the long-run mean $\mu = P^* = \frac{\gamma}{\omega}$, which is exactly the equilibrium price derived from the deterministic model. Recalling that $\alpha_P = \alpha/\omega$ and $\alpha = t/\beta$, the Price function composed with $\alpha_P(t)$ is:

$$P(\alpha_P(t)) = P^* - Ce^{-t/\beta}$$

The OU process therefore recovers this as its expected path:

$$\mathbb{E}[P(\alpha_P(t))] = P^* + (P_0 - P^*)\,e^{-t/\beta}$$

The stochastic extension thus adds a variance structure around the deterministic solution without altering its mean behaviour [Uhlenbeck and Ornstein, 1930].

An analogous extension applies to the Quantity-Beta ODE:

$$\mathrm{d}Q = \underbrace{\frac{1}{\beta}\left(\frac{\delta}{\varepsilon} - Q\right)\mathrm{d}t}_{\text{deterministic (mean-reversion)}} + \underbrace{\sigma_Q\,\mathrm{d}W_t^Q}_{\text{stochastic (noise)}}$$

where W_t^Q may in general be correlated with W_t^P, reflecting the coupling between price and quantity fluctuations in a real market.

8 Conclusion and Why Economics is a Human Science

My Free-Fall ODE was tested against Smith's Test 1 Smith [1962]. There indeed existed a β for which the Price-Beta function $P_\beta(t) = P_\alpha(t/\beta)$ mapped reasonably well onto the t domain. Although the fit was not perfect, the model captured the main convergence trend toward the equilibrium price. This was further supported by the Monte Carlo CDA experiment of §6.2.2: across 750 simulations spanning five qualitatively distinct supply–demand geometries (symmetric linear, asymmetric linear, flat-supply, flat-demand, and nonlinear logistic/quadratic-decay) and two initial-condition regimes, realized contract prices clustered near the schedule intersection P^*, exactly so for symmetric schedules and with a small predictable offset for asymmetric and nonlinear schedules. This suggests that the first-order relaxation framework is not merely algebraic, but is also capable of describing the qualitative motion of an idealized Market toward Equilibrium under a range of microstructural realizations and across both linear and nonlinear monotonic schedules.

Even though experimental Quantity results were not available, the Quantity-Alpha function $Q_\alpha(\alpha)$ was derived consistently within the model by composing Q_{supply} with $P_\alpha(\alpha)$ in the shortage regime. Therefore, the Quantity-Beta function

$$Q_\beta(t) = Q_\alpha(t/\beta)$$

inherits the same theoretical time-scaling as the Price-Beta function

$$P_\beta(t) = P_\alpha(t/\beta).$$

However, since Quantity data was not directly available from Smith's Test 1, this Quantity path should be understood as a model-implied trajectory rather than an empirically verified one.

I believe I have shown how alike an idealized Market and a Physical System can be. As an Engineer, I find it absolutely beautiful. However, it must be noted that the entire reason why the solutions were mathematically "nice" is the assumption of *Ceteris Paribus*. This assumption holds external factors fixed, allowing Demand and Supply to be treated as clean functions of Price and Quantity. Under the linear schedules used in this paper, every relevant Quantity corresponds to both a Demand Price and a Supply Price, and every relevant Price corresponds to both a Demand Quantity and a Supply Quantity. Therefore, the net market imbalance is defined along the adjustment path, allowing the model to assign a direction of motion toward the unique Equilibrium where P_{supply} and P_{demand} are equal.

This is where surjection has a meaningful role in the restricted model. The Supply and Demand relations are taken to cover the relevant admissible domain of market states, so there is no point along the considered path where Demand or Supply becomes undefined. Because of this, the Market always has a well-defined imbalance, and if there is an imbalance, there is a

corresponding direction of motion. This is what makes the First-Order relaxation treatment possible.

In real cases, however, the state of the Market cannot be determined so easily. In fact, it perhaps can never be determined completely. The state of a Market is the result of human emotions, incomplete information, expectations, delays, external shocks, and several other variables. This makes the Market practically impossible to model as a purely deterministic system. Deterministic Models, such as the one used in this article, can describe idealized aspects of Market behaviour, but they cannot fully govern or predict a system driven by human decision-making.

Therefore, Economics, the study of Markets driven by human behaviour, cannot be viewed purely through a mathematical lens. Mathematics can describe important aspects of it, but mathematics cannot fully determine it. This is what makes Economics a Human Science. In being a Human Science, Economics can resemble a Physical System under idealized assumptions, but it will never be a fully Deterministic Physical System in reality.

However, Statisticians, Economists, and Engineers still use mathematics and physics to estimate the state of a Market based on historical data. This is largely seen in Quantitative Finance, where firms trade assets using Probabilistic models, Stochastic Processes, and Stochastic Calculus. Brownian Motion, which originated in physics as a model of random particle motion, appears in finance as a model of random Market fluctuations. Similarly, the Ornstein–Uhlenbeck process extends First-Order mean-reverting dynamics by adding stochastic noise, showing how the deterministic relaxation structure studied in this paper naturally connects to accepted stochastic models.

The bottom line is that the Economic Market is beautiful because it sits between order and randomness. Under *Ceteris Paribus* and other idealized assumptions, it can move like a Physical System toward Equilibrium. In reality, because it is driven by human behaviour, it remains complex, noisy, and only partially predictable.

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